Family Resources and Human Capital in Economic Downturns

by

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CES 24-15 **March 2024**

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Abstract

I study how recessions impact the human capital of young adults and how these effects vary over the parent income gradient. Using a novel confidential linked survey dataset from U.S. Census, I document that the negative effects of worse local unemployment shocks on educational attainment are strongly concentrated among middle-class children, with losses in parental home equity being potentially important mechanisms. To probe the aggregate implications of these findings and assess policy implications, I develop a model of selection into college and life-cycle earnings that comprises endogenous parental transfers for education, multiple schooling options, and uncertainty in post-graduation employment outcomes. Simulating a recession in the model produces a "hollowing out the middle" in lifecycle earnings in the aggregate, and educational borrowing constraints play a key role in this result. Counterfactual policies to expand college access in response to the recession can mitigate these effects but struggle to be cost effective.

[*] I am heavily indebted to John Kennan for his insights and patience. Thanks also to Dean Corbae, Diego Daruich, Joel McMurry, Jo Mullins, Mike Ricks, Ananth Seshadri, Jeff Smith, Brittany Street, Joanna Venator, Chris Taber, Sarah Turner, Matt Wiswall, Anson Zhou, and audiences at SOLE and SED for many helpful comments. Thanks to Bob Thomas for every step of applying for and using the restricted-use ACS and Decennial data. This research was partially supported by the Summer Research Fellowship generously provided by the University of Wisconsin-Madison. Additionally, this material is based upon work supported by the National Science Foundation Graduate Research Fellowship Program under Grant No. DGE-1747503. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author and do not necessarily reflect the views of the National Science Foundation. Support was also provided by the Graduate School and the Office of the Vice Chancellor for Research and Graduate Education at the University of Wisconsin-Madison with funding from the Wisconsin Alumni Research Foundation. Any views expressed are those of the authors and not those of the U.S. Census Bureau. The Census Bureau's Disclosure Review Board and Disclosure Avoidance Officers have reviewed this information product for unauthorized disclosure of confidential information and have approved the disclosure avoidance practices applied to this release. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2413. (CBDRB-FY20-P2413-R8725 and CBDRB-FY22-P2413-R9572). All remaining errors are my own.

1 Introduction

How do recessions impact human capital accumulation among young adults, and how do these effects vary based on the resources of one's family? Economic downturns decrease the opportunity cost of spending more time building human capital as opposed to working, such as by going to college, but also often feature decreases in family resources through events such as layoffs or asset depreciation. Entries in the economics literature have documented that early exposure to recessions can have long-term negative effects on earnings (Kahn, 2010; Oreopoulos et al., 2012; Stuart, 2022), as well as that the role of parental resources and educational borrowing constraints in determining college attendance has increased in recent decades (Lochner and Monge-Naranjo, 2011). How these forces interact, however, is not well known and has important implications for understanding the incidence of recessions and informing policy responses to them.

This paper studies the role of family resources in human capital accumulation and earnings in the context of the financial crisis of 2007-08 and the subsequent Great Recession. The financial crash induced severe shocks to employment and housing wealth across the country, with the intensity of these shocks varying considerably across labor markets. This spatial heterogeneity has allowed researchers to use US local labor markets as a laboratory in which to study the effects of the Recession (Yagan, 2018). I first conduct an empirical investigation that uses a novel dataset comprised of linked confidential data from U.S. Census. New linkage rules and methods allow for individuals in the yearly American Community Surveys to be linked to themselves in the 2000 Census — in addition to providing the likely local labor market of these individuals at the time of the financial crash, this also allows me to obtain information about their parents and family background if they were a child in 2000. I focus my analysis on the birth cohorts of 1990 through 1994, as these individuals would have been age 13-17 in 2007 and thus were at considerable risk of entering the labor force in a bad economy.

Exploiting the aforementioned spatial variation in shock severity across commuting zones (CZs) and individual variation in parental resources, I find evidence that college enrollment and attainment is considerably less countercyclical for middle-class children than it is for either their poor or rich counterparts. Other things held equal, exposure to a one percentage point larger 2007-2009 CZ unemployment shock in the recession is associated with a 7% long-run reduction in college attainment for children with parents in either the second or third quartile in the parental income distribution. The effects for children with bottom

and top-quartile parents are a 1% gain or 2% reduction respectively, with neither effect being statistically significant. Higher rates of college non-completion appear to be more important in generating this finding than substitution to two-year degrees, and additional evidence points to losses in parental home equity being a particularly important mechanism for reducing college attainment among middle-class children.

The data, however, are limited in their ability to speak to lifecycle and aggregate implications of the recession. To probe these issues and study policy implications, I develop and estimate a model of selection into college and life-cycle earnings. The model most closely resembles that of Colas et al. (2021) in featuring endogenous parental transfers, multiple schooling outcomes, and post-graduation labor market uncertainty. The novel features of my model include multiple schooling options (allowing agents to potentially substitute into cheaper but lower-quality options in response to family resource shocks) and post-graduation uncertainty in *employment* as well as earnings. Simulating a shock resembling the Great Recession results in a "hollowing out of the middle" in lifecycle earnings in the aggregate that mirrors the partial effect found in the reduced-form section: while the recession barely alters the gap in lifetime earnings between top and bottom-decile children, the gap between middle and top-decile children rises by $60,000. Educational borrowing constraints play a key role in generating this result, with the hollowing effect of the recession vanishing entirely in a setting with no student loan limits. Other counterfactual policies intended to ease college access in response to the downturn can also mitigate these effects but generally fail to be cost-effective due to parental crowd-out and substitution into lower-quality college options during a contraction.

Related Literature

This paper contributes to several strands of economic literature, the first being the one that studies the effects of recessions on individual outcomes. Exposure to recessions has long-lasting consequences for earnings and human capital (Stuart, 2022), with particularly large effects for workers who were displaced during an economic downturn (Davis and Wachter, 2011). Yagan (2018) finds that the local shocks induced by the Great Recession had large effects on the employment and income of working-aged adults. Rinz (2022) finds that adolescents in the United States were even more strongly affected by the crash in percentage terms, and Bell and Blanchflower (2011) reach the same conclusion for youth in the United Kingdom. However, neither study is able to explore heterogeneity in effects across parental socioeconomic status in their samples. Moving from purely economic impacts, (Giuliano and Spilimbergo, 2013) find that youth exposed to economic downturns exhibit different

redistribution preferences than non-exposed peers.

Past literature has also established that entering the labor force in a recession has long-lasting negative impacts on individual earnings (Kahn, 2010; Oreopoulos et al., 2012). However, the extent to which individuals attempt to avoid or mitigate the adverse effects of recessions is not well understood. Yagan (2014) finds that working-age adults did not appear to move to less-affected labor markets in response to local economic shocks. Long (2014) reports that college attendance increased more in states that were more affected by the recession, indicating some evasive action being taken on the part of adolescents; (Charles et al., 2018) also find that college enrollment moved opposite with the housing boom and bust in the 2000s. Focusing the heterogeneity of effects of recessions over the parent resource gradient is my main contribution to this literature.

In doing this, this paper also speaks to the large literature that has studied the role of family resources and credit constraints in college enrollment and completion and extends it through studying the interaction of these factors with the business cycle. While older entries in this literature suggest a limited role of family income in deciding whether children go to college (Cameron and Taber, 2004), other papers have documented that family resources may have become more important in recent years (Belley and Lochner, 2007; Lochner and Monge-Naranjo, 2011). This empirical results in this paper provide support for the latter interpretation and suggest different roles for family income and family wealth in financing college enrollment, offering a potential resolution between papers that have separately documented small parental income effects on college enrollment (Hilger, 2016) but large effects for parental home equity (Lovenheim, 2011).

The third literature I contribute to is the one that models human capital formation in earlier parts of the lifecycle. Parent inputs and resources are vital in early (Becker and Tomes, 1979; Cunha and Heckman, 2007) and, increasingly, later stages of child human capital development — this paper complements these papers by studying the importance of parental inputs and resources on the transition to adulthood, which Danziger and Ratner (2010) argue has become considerably more difficult over recent years. The model I develop most closely resembles the models in Heckman et al. (1998) and Colas et al. (2021), with the main contribution being that I enhance the realism of educational decisions by explicitly including a richer menu of college enrollment decisions and include employment risk after graduation. Multiple college options is a particularly salient feature, since substitution to lower-quality options can have marked impacts on later-life outcomes (Mountjoy, 2022). In evaluating the incidence of counterfactual tuition and student loan policies, I also contribute

to the literature that has studied the optimal design of such policies (Johnson, 2013; Abbott et al., 2019) by providing estimates regarding how it varies over the business cycle.

Methodologically, this paper is one of the first to form intergenerational linkages *across* mandatory 21st-century Census surveys. Individuals in older vintages of Census data have been linked across time via probabilistic matching algorithms in past research (e.g. Ferrie (1996); Feigenbaum (2016)), but issues of match quality produced by these algorithms can be severe (Bailey et al., 2020). The use of administrative records and the finer geographic information available in restricted-use Census survey data allows for cross-survey linkages to be made considerably more confidently and frequently, which may make the data I use a valuable resource for researchers going forward.

The remainder of the paper proceeds as follows. Section 2 introduces the data I use in this paper along with my empirical strategy, while Section 3 details my empirical results. Section 4 lays out my model, and Sections 5 describes my estimation strategy and evaluates the model's fit. Sections 6 studies the effects of the Recession in the model environment and its interaction with counterfactual policy. Section 7 concludes.

2 Data

2.1 Sample Description

To estimate how the effects of the Great Recession on human capital accumulation among young adults varied over family resources, I use newly available linked confidential data consisting of the 2000 Decennial Census and the 2005-2018 waves of the American Community Surveys. In addition to containing considerably richer geographic information for the respondents, individuals in both these datasets have been linked to the Social Security Administration NUMIDENT file using the U.S. Census Bureau's Person Identification Validation System, which allows me to assign Protected Identification Keys (PIKs; essentially scrambled SSNs) to the large majority of individuals in the surveys[1].

While other papers have made use of these data as repeated cross-sections (see for instance Stuart (2022)), recent regulation changes regarding linking mandatory Census surveys allow

[1]The PVS is Census Bureau's system for searching and verifying Social Security Numbers or Protected Identification Keys for person records in demographic surveys and/or censuses. This system first attempts to match individuals in surveys to SSA records by using address information — if this fails, individuals are matched via name and date of birth. A more comprehensive review of this procedure and its performance can be found in Mulrow et al. (2011)

Table 1: ACS Linkage Rates

Dataset	PIK Assignment Rate	Census Link Rate
2005 ACS	90.5	10.0
2006 ACS	90.5	16.6
2007 ACS	90.3	15.4
2008 ACS	90.2	14.4
2009 ACS	89.6	14.0
2010 ACS	93.1	10.7
2011 ACS	91.9	15.2
2012 ACS	91.6	13.9
2013 ACS	91.6	13.1
2014 ACS	91.9	12.8
2015 ACS	91.9	10.9
2016 ACS	91.5	13.3
2017 ACS	91.9	12.7
2018 ACS	93.0	12.6

Notes: Data from 2000 Census and 2005-2018 American Community Surveys. Statistics are for 1990-1994 birth cohorts. Linkage rates rounded for disclosure avoidance purposes. Column 2 reports the rates at which individuals in the relevant year of ACS data are assigned PIKs. Column 3 reports the rate at which individuals in the relevant year of ACS data are linked to themselves in the 2000 Long-Form Census.

for respondents in these surveys to be *linked to themselves* over time[2]. With PIKs being stable over time and the large majority of U.S. citizens appearing in either the short or long-form 2000 Census, this allows me to link individuals in the ACS to themselves in the 2000 Census, effectively forming a two-year panel for a 1% sample of the U.S. over the years 2005 to 2018. This linkage allows me to observe detailed family background information among adults in the ACS who no longer live with their parents, overcoming a key limitation in the public-use version of the data.

My baseline sample consists of individuals born between 1990 and 1994 in any ACS year. These birth years translate to these cohorts being aged between 13 and 17 in 2007 — thus, these cohorts would have been at considerable risk of entering the labor force during the recession. Further, because these birth cohorts were children in the year 2000, linking these individuals in the ACS to themselves in the 2000 Census allows me to infer their family background by observing information about household heads and their spouses[3].

[2]Conversations with Census Research Data Center administrators suggest that this change happened sometime in the 2019 calendar year.

[3]Over 90% of individuals in these birth cohorts are coded to be either the child, stepchild, or adopted

For individuals linked to themselves in the 2000 short form, I can see their parents' race, marital status, age, and location down to the census block. For those linked to the long-form Census, I can see the same information along with a rich vector of parent demographic and socioeconomic covariates. I limit the sample to individuals who were observed in the long-form Census and were coded as either the child, adopted child, or stepchild of the household head. Individuals living in Group Quarters in the 2000 Census are dropped, as are any groups of individuals in any survey year who are linked to the same PIK.

After performing the linkages and applying the sample restrictions listed above, I am left with slightly over 500,000 observations of the 1990-1994 birth cohorts over the 2005-2018 period. Table 1 reports the rates at which I can both assign individuals in the ACS PIKs and link them to themselves in the 2000 Census. Broadly speaking, the linkage is a success: I am able to obtain family background information for more than 12% of the relevant individuals in most of my ACS years, which is reasonably high given that only one sixth of households filled out the 2000 Long-Form Census in the first place. Table 2 reports how basic demographic and socioeconomic characteristics of my ACS sample change after removing unlinked individuals in either step. The table suggests that the linkage process does induce some selection, with the economic indicators of the sample improving somewhat after the linkage procedure; in particular, many blacks are dropped in the linkage procedure. This is not surprising, as the PIK system relies most heavily on individual addresses to make linkages; thus, individuals who do not have stable addresses are more likely to be unmatched and subsequently dropped. To address this selection issue, I calculate the ex-ante probability of being included in the baseline sample by state, gender, and race (white, black, Hispanic or Asian) and modify individual sample weights by dividing them by these probabilities[4].

2.2 Empirical Strategy

I estimate the effects of exposure to the Great Recession using an event-study specification that is similar to Yagan (2018) and Rinz (2022). Specifically, I estimate the following regression:

$$y_{iact} = \boldsymbol{\alpha} \boldsymbol{X}_i + \theta_c + \gamma_a + \delta_t + \beta_a S_c + \varepsilon_{iact}, \tag{1}$$

child of their household head in 2000.

[4]In practice, using these modified weights vs. the original sample weights does not change the sign or significance of the empirical results.

Table 2: Sample Summary Statistics

Variable	ACS	ACS (PIK Linked)	ACS (Census Linked)
Age	19.07	19.08	19.03
	(4.36)	(4.36)	(4.32)
Male	50.92	50.74	50.86
	(49.99)	(49.99)	(49.99)
White	59.64	61.47	71.92
	(49.06)	(48.67)	(44.94)
Black	15.54	15.37	11.31
	(36.23)	(36.07)	(31.67)
Asian	4.85	4.58	2.78
	(21.49)	(20.89)	(16.44)
Hispanic	21.10	19.70	14.71
	(40.80)	(39.78)	(35.42)
Married	6.99	6.95	6.81
	(25.50)	(25.43)	(25.20)
Employed	41.34	41.98	43.15
	(49.24)	(49.35)	(49.53)
Wage Income ($1,000)	9.85	9.97	10.21
	(16.27)	(16.33)	(16.41)
College Degree	9.47	9.71	10.14
	(29.28)	(29.60)	(30.18)
Associate Degree	13.12	13.48	14.17
	(33.76)	(34.15)	(34.88)
High School Degree	52.85	53.22	53.21
	(49.92)	(49.90)	(49.90)
In Northeast	17.11	17.23	17.35
	(37.66)	(37.76)	(37.87)
In Midwest	21.33	21.95	26.83
	(40.96)	(41.39)	(44.31)
In Southeast	36.99	36.91	34.52
	(48.28)	(48.26)	(47.54)
In West	24.57	23.91	21.29
	(43.05)	(42.65)	(40.94)
N	4053000	3702000	540000

Notes: Data from 2000 Census and 2005-2018 American Community Surveys. Statistics are for 1990-1994 birth cohorts. Standard deviations in parenthesis. Sample sizes rounded for disclosure avoidance purposes. Column 2 reports variable means and standard deviations for entire ACS sample. Column 3 reports variable means and standard deviations for ACS sample linked to PIKs. Column 4 reports variable means and standard deviations for ACS sample linked to 2000 Long-Form Census.

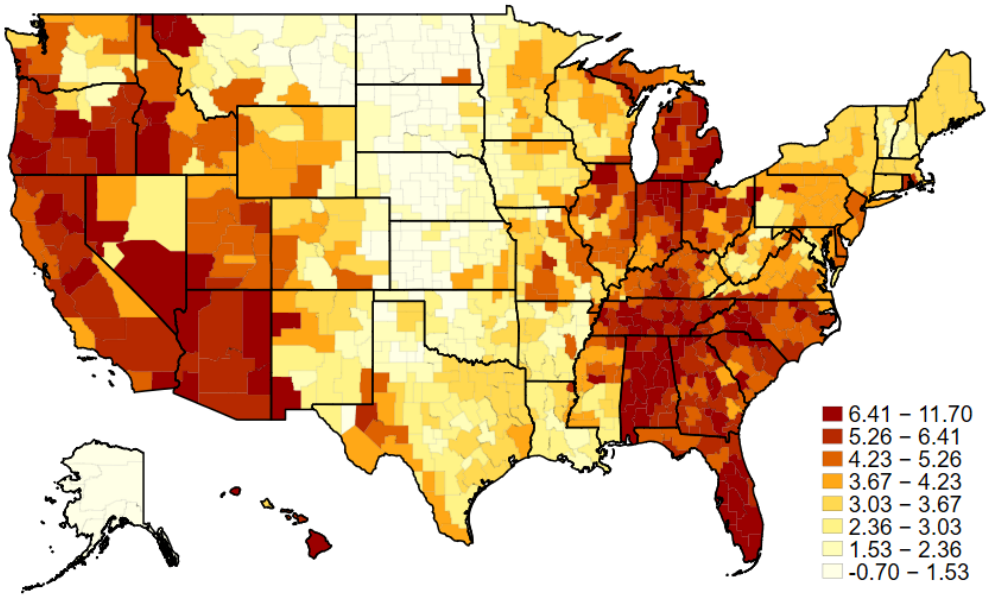
where y_{iact} is some outcome variable of interest (such as earnings or educational attainment) for individual i at age a who was in commuting zone (CZ))[5] c in 2007 and surveyed in year t. Fixed effects for age, CZ, and calendar year $\gamma_a, \theta_c, \delta_t$ control for life-cycle patterns, geographic mean differences, and aggregate yearly trends in the outcome variable, and $\boldsymbol{X}_i$ is a rich vector of demographic and background covariates including parental marital status, parental income, parental homeownership, parental house value, parental education, and a dummy for the year-2000 household head working in either construction or manufacturing. The error term is represented by ε_{iact}. Following Yagan (2018), S_c is the 2007-2009 change in unemployment rate observed in commuting zone c. Thus, the β_a terms are the coefficients of interest and have the interpretation of the effect on some outcome at age a of living in 2007[6] in a labor market that experienced a one percentage point larger unemployment shock in the Great Recession.

To facilitate the inclusion of the CZ fixed effects θ_c, I normalize one of the β_a terms to be zero. The identifying assumption in this specification is parallel trends, or children in CZs that received larger shocks in the recession were not systematically trending differently in the outcome variable compared to children in CZs that received smaller shocks. Unfortunately, the focus on individuals who were still attached to their parents in 2007 results in pre-trends for outcomes such as earnings and college attainment being largely unviewable, so the identifying assumption is difficult to verify directly and is tantamount to selection on observables. I aim to make the identifying assumption credible by including a comprehensive set of controls in my regressions — additionally, I verify that the empirical specification applied to the entire sample can reproduce estimates of Great Recession earnings impacts reported in other entries of the literature as a face validity test. I cluster standard errors by year-2000 CZ to allow for arbitrary serial correlation of error terms within CZs.

[5]Commuting Zones are a frequently used notion of local labor markets and groups of counties characterized by strong economic ties. There are 722 CZs in the United States. CZ shocks are computed by taking county-level unemployment rates from the monthly BLS Local Area Unemployment Statistics series and then aggregating to CZs with population weights before averaging across months to form a yearly CZ measure. For a visualization of these shocks, refer to Figure 1.

[6]One limitation of my data is that the 2007 CZ of residence for survey respondents is not directly observable, so I use their year-2000 CZ of residence as a proxy. Looking at the 2007 ACS allows a direct evaluation of how good this approximation is: I find that 87% percent of my sample in 2007 lived in the same CZ as in 2000, with minimal heterogeneity over the parent income gradient.

Figure 1: Great Recession CZ Shocks



Notes: Data from BLS LAUS series.

3 Empirical Results

3.1 Results from Full Sample

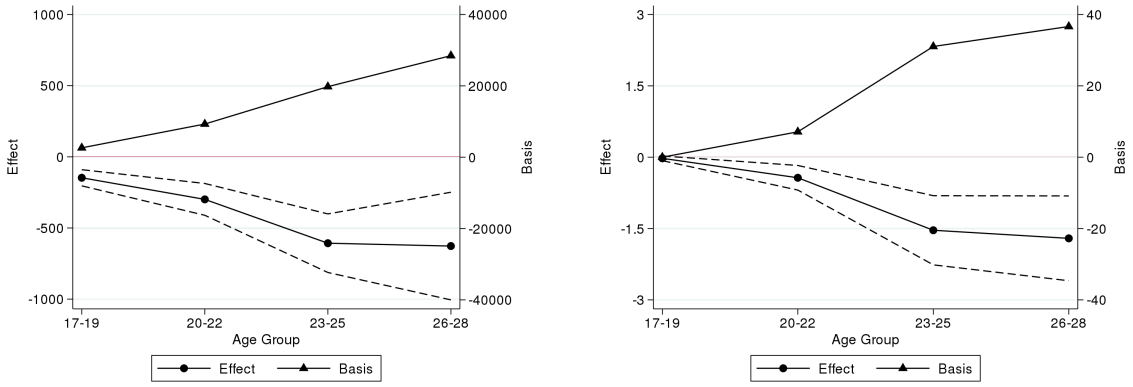
I first estimate the effects of the recession on earnings and human capital for my entire sample to assess the face validity of the output of my specification. When estimating (1), I group ages into five bins ($< 17, 17-19, 20-22, 23-25, 26-28$) to increase precision and to reduce disclosure risks. The first age group is used as the omitted category, so all estimated coefficients for a given outcome variable are interpretable as the effects of the Recession relative to its impacts on individuals aged younger than 17. Effects on binary variables are converted to percentage points terms to ease interpretability.

Figure 2 displays the estimates of the effects of the Great Recession on wage earnings and college attainment over age groups[7]. The estimates suggest that exposure to a 1 pp larger unemployment shock in 2007 is associated with a reduction of $600 in annual earnings and a 1.7 pp lower probability of college attainment or higher by age 26-28.

Pre-trends are difficult to observe for these outcome variables since few individuals were

[7]For effects on attainment of at least an associate degree or some college, see Figure A.1.

Figure 2: Full-Sample Effects of the Recession on Earnings and College Attainment



(a) Wage Earnings (b) College Attainment

Notes: Figures display outcome variable means over age and estimates of β_a from equation (1), with age grouped into five bins and effects for the bottom age group normalized to zero. Outcome variables include annual earnings from wages/salary and a dummy variable for attainment of at least a four-year college degree. Standard errors are clustered at the year-2000 CZ level, and the dotted lines indicate 95% confidence intervals for coefficient estimates. Coefficient estimates correspond to the circular dots and are measured by the left-hand axes; bases correspond to triangular dots and are measured by the right-hand axes. The sample includes individuals of the 1990-1994 birth cohorts observed in the 2005-2018 ACS and linked to themselves in the 2000 Long-Form Census; see text for details.

earning income and none had attained a college degree before age 17 — however, other papers that have used the same CZ unemployment shock variation allow me to assess the face validity of these estimates. Yagan (2018) finds that exposure to a 1pp larger shock for individuals who were age 25-54 is associated with a long-term reduction in earnings of $1,000; Rinz (2022) replicates this result and conducts subgroup analyses of the effects of the Recession on earnings over different generations (Millenials, Gen-X, Baby Boomers, and the Silent Generation). He estimates a long-run effect of exposure to a 1 pp larger shock of approximately $600 for Millenials, suggesting that the effects I estimate are quite reasonable. Rinz (2022) also estimates a 1.3pp reduction of completion of any post-secondary degree, which is very similar to my estimate for the effects of the Recession on college attainment. That my estimate is somewhat larger is reasonable that I focus on college attainment of the attainment of any postsecondary degree — additionally, his Millenial sample contains individuals who were born at 1981 at the earliest and thus would have contained a substantial number of individuals who already completed their education by the time the Great Recession

occurred. The results for college attainment are also similar in magnitude to those reported in Stuart (2022), who finds that a one standard deviation[8] increase in exposure to the 1980 recession is associated with a 3 pp decrease in four-year college attainment.

3.2 Parent Resources and the Effects of the Recession

The results using the full sample provide evidence for the validity of my empirical specification. To study how the effects of the Great Recession on human capital accumulation and earnings varied over the parent income gradient, I conduct subgroup analyses by estimating (1) over parent income quartiles[9]. I prefer subgroup analyses to including interaction terms in the regression, as the former allows for age/year fixed effects and the demographic controls in $\boldsymbol{X}_{it}$ to differentially impact the outcome variable of interest based on socioeconomic background in a fully flexible way.

Figure 3 presents the results of this exercise for four-year college attainment or higher, with individual plots structured similarly to those in Figure 2. By age 26-28, at which point college attainment rates flatten considerably in the data, exposure to a 1pp larger unemployment shock in the Great Recession is associated with a 2pp or larger reduction in four-year college degree attainment for children who had parents in the second or third income quartiles. The comparable estimate for children with top-quartile parents is smaller (slightly over 1pp) and not statistically distinguishable from zero, though equivalence of the coefficients for second/third-quartile children is difficult to reject. However, the estimate age 26-28 coefficient for bottom-quartile children is essentially zero and is statistically distinguishable from the second/third quartile effects.

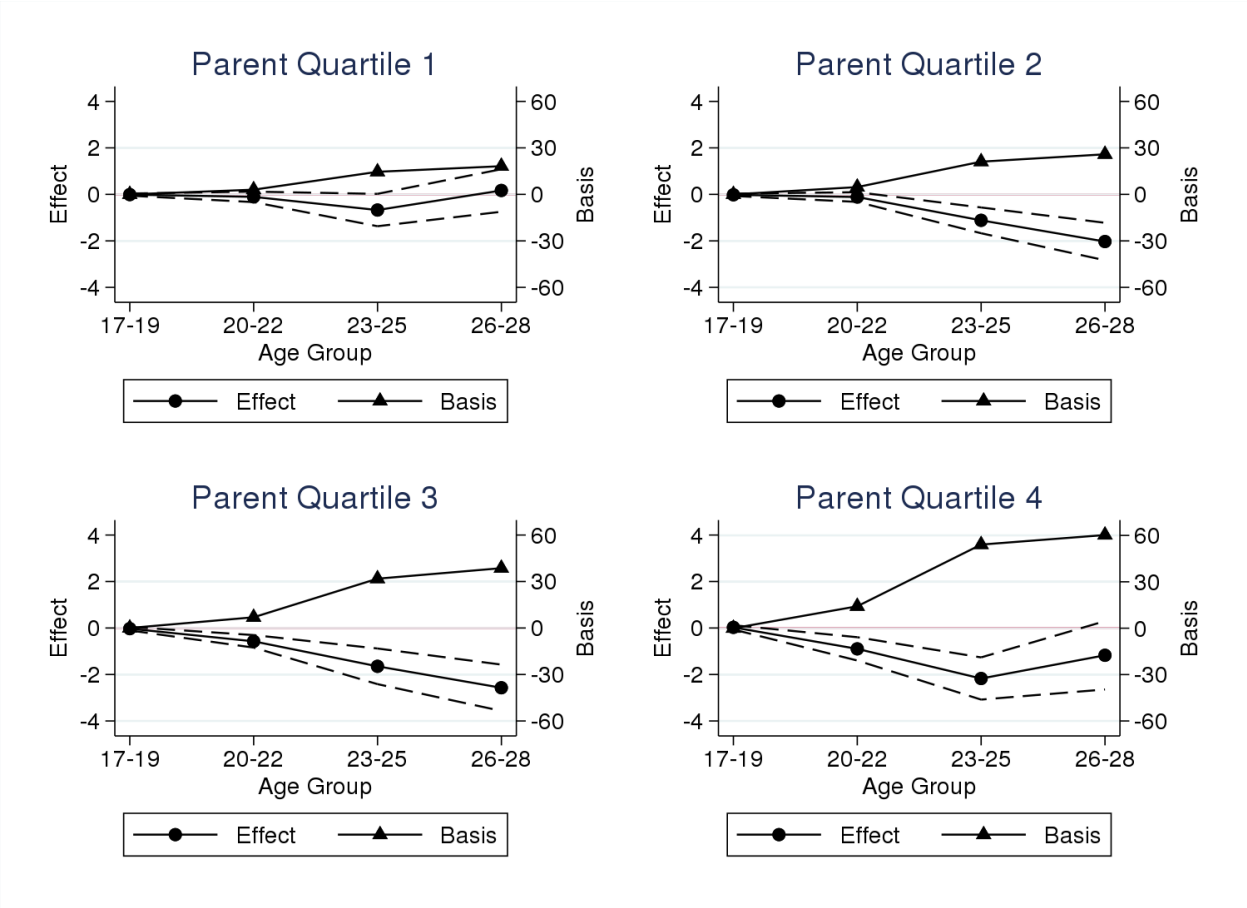
Also of interest is that children from all types of socioeconomic backgrounds see drops in college attainment at age 23-25. The null effects for top and bottom children come from a recovery in college attainment as opposed to null effects over the entire life cycle. This suggests that the recession may have *delayed* four-year college degree attainment for all types of children but that these delays became permanent primarily for middle-class children while wealthier children became more likely to pursue non-traditional education paths[10].

[8] A 1pp larger CZ unemployment shock is around a 0.65 SD increase in my specification.

[9] Parent income is defined as the sum of household head and spouse of household head (if present) wage income in 2000. Quartiles are computed within-cohort to account for potential age differences of parents for individuals who were born in earlier or later years.

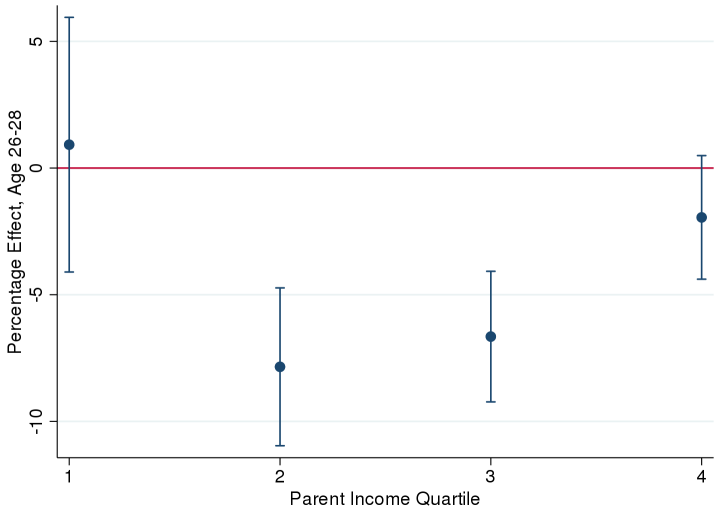
[10] Figure A.2 appears to corroborate this interpretation, with top-income children being more likely than others to be enrolled in college at age 23-25.

Figure 3: Effects of Recession Shocks on College Attainment by Parent Income Quartile



Notes: Figures display outcome variable means over age and estimates of β_a from equation (1) for the outcome variable of at attainment of at least a four-year college degree, with age grouped into five bins and effects for the bottom age group normalized to zero. Effects estimated separately over year-2000 parent income quartile. Standard errors are clustered at the year-2000 CZ level, and the dotted lines indicate 95% confidence intervals for coefficient estimates. Coefficient estimates correspond to the circular dots and are measured by the left-hand axes; bases correspond to triangular dots and are measured by the right-hand axes. The sample includes individuals of the 1990-1994 birth cohorts observed in the 2005-2018 ACS and linked to themselves in the 2000 Long-Form Census; see text for details.

Figure 4: Percentage Effects for College Attainment at Age 26-28 Over Parent Income Quartile

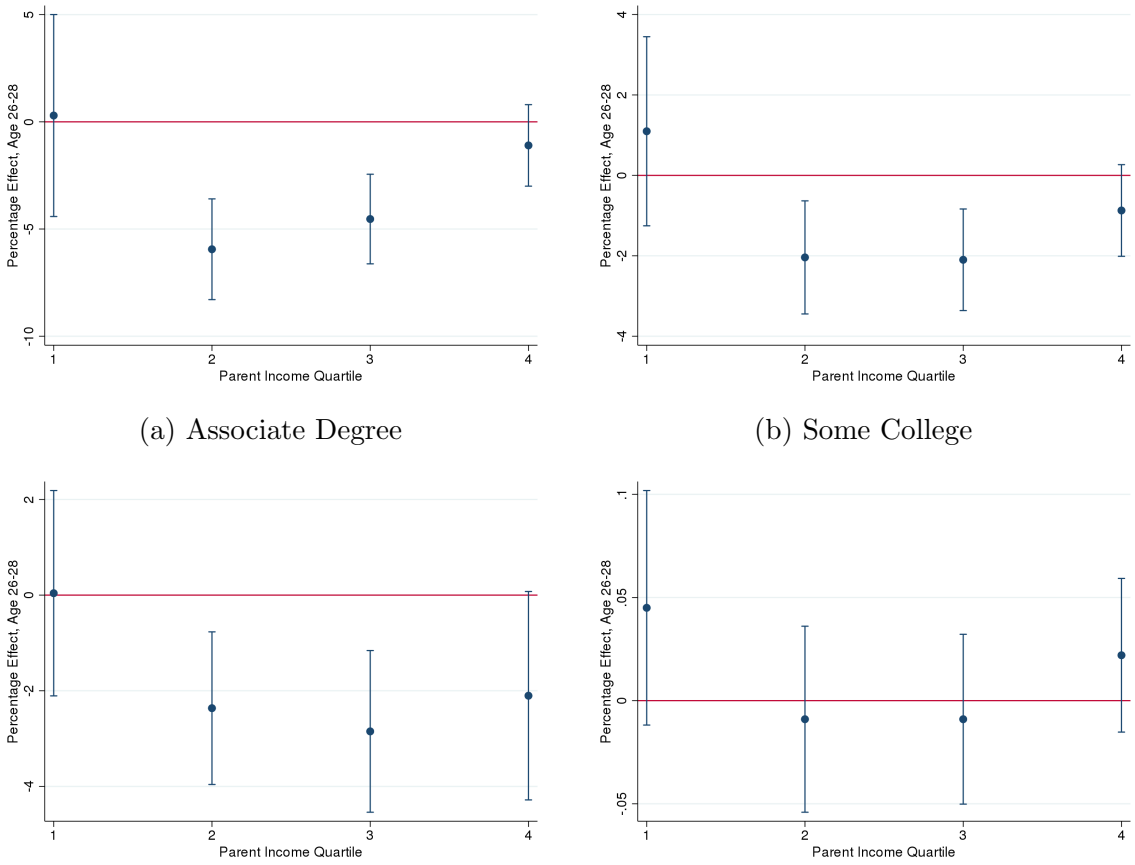


Notes: Figures display estimates of β_{26-28} from equation (1) for the outcome variable of attainment of at least a four-year college degree over parent income quartile. Coefficient and standard error divided by outcome variable mean for relevant parent quartile group at age 26-28 to form percentage effects. Standard errors are clustered at the year-2000 CZ level, and 95% confidence intervals are displayed. The sample includes individuals of the 1990-1994 birth cohorts observed in the 2005-2018 ACS and linked to themselves in the 2000 Long-Form Census; see text for details.

An important caveat to consider is that the effects displayed in Figure 3 are in levels as opposed to percentage terms, which is relevant because the base rates of college completion vary substantially over the parent income gradient. Roughly 30% of children with second or third-quartile parents had completed a four-year college degree or higher by age 26-28, while the corresponding figure for top-quartile children was over 60%. Figure 4 takes the estimates and standard errors of β_{26-28}[11] for the four types of children and divides them by the bases to obtain percentage effects before plotting these effects over parent quartiles. A clear pattern emerges: in percentage terms, long-term human capital losses from the Great Recession were sharply concentrated among children from the middle class. Exposure to a 1pp larger shock is associated with a roughly 7% decline in college attainment for these children, while top/bottom-quartile children saw an insignificant 2% loss and 1% gain[12].

[11] One can interact the CZ shocks with time indicators instead and obtain similar results; see Figure A.3.
[12] Discretizing these impacts over income deciles reveals a comparable U-shape; see Figure A.4.

Figure 5: Percentage Effects at Age 26-28 Over Parent Income Quartile



(a) Associate Degree

(b) Some College

(c) Wage Earnings

(d) Log Wage Earnings

Notes: Figures display estimates of β_{26-28} from equation (1) for the indicated outcome variable over parent income quartile. Coefficient and standard error divided by outcome variable mean (except for log wage earnings) for relevant parent quartile group at age 26-28 to form percentage effects. Standard errors are clustered at the year-2000 CZ level, and 95% confidence intervals are displayed. The sample includes individuals of the 1990-1994 birth cohorts observed in the 2005-2018 ACS and linked to themselves in the 2000 Long-Form Census; see text for details.

Figure 5 conducts the same exercise for alternate outcome variables: Figure 5a considers attainment of an associate degree or higher, Figure 5b some college or higher, and Figures 5c and 5d evaluate the percentage/log effects for wage earnings[13]. A very similar U-shaped pattern of heterogeneity emerges when evaluating the effects on associate degree attainment, though the impact on middle-class children attenuates to be around 5%. When considering some college attainment or higher, the effects attenuate markedly further, and the U-shape pattern flattens considerably, though the percentage effects for top/bottom-quartile children are still different from those of middle-quartile children at a $p < 0.05$ level. This suggests that college non-completion was a more important mechanism than substitution into two-year degrees in generating the results plotted in Figure 4, consistent with evidence that enrollment expansions associated with recessions are concentrated among lower-quality colleges (Bound et al., 2010; Barr and Turner, 2013).

The U-shape heterogeneity persists when looking at wage earnings and log wage earnings, though pattern does become somewhat noisier. Figure 5c demonstrates that long-run percentage effects on wage earnings from the Great Recession were only statistically distinguishable from zero for middle-class children, though equivalence of the coefficients for middle-class and top-quartile children clearly cannot be rejected. However, the effects for bottom-quartile children, in addition to being almost exactly zero, are statistically distinguishable from those of other types of children. This provides evidence that shocks to human capital accumulation via schooling were a more important mechanism for decreasing long-term earnings than was potential human capital depreciation during unemployment, as children of all types saw decreased employment following the Recession at first (see Figure A.5). If human capital depreciation during unemployment in the Recession was severe, we may expect to see long-term earnings reductions for low-income children as well.

In Figure 5d I discard individuals with zero earnings and consider the effects of the Recession on log wages. In this case, none of the long-term effects are distinguishable from zero, suggesting that reduced employment was the dominant mechanism for reducing earnings for these cohorts. Among the employed, however, the estimated effects for top and bottom-quartile children are statistically distinguishable from the effects for middle-income children at the $p < 0.10$ level, likely reflecting the differential effects the Recession had on their human capital accumulation. That the heterogeneity I document for human capital accumulation is sharper than it is for earnings is not surprising given that the college earnings premium is increasing over the life cycle. Since college attainment affects one's earnings trajectory as

[13]Level plots of these outcome variables are available upon request.

well as earnings level, the effects of decreased college-going on earnings are likely to not be nearly as strong at age 26-28 as they will be at age 40 or later. Enabling the study of the full life-cycle earnings impacts of the Great Recession is a key motivator for building the model I present in the subsequent section.

3.3 Mechanisms

The previous results provide strong evidence that the Great Recession decreased human capital attainment for children from middle-class families. In this section, I investigate the mechanisms that drove these impacts. I consider two potential mechanisms in particular that may have made college attendance prohibitively expensive for these children: losses in parental income following layoffs in the Recession, or losses in parental house equity following the housing market crash. To study these possible explanations, I (run 1)for children with second or third-quartile parents broken down by whether their parents lived in a house with positive value in 2000 and whether the household head worked in either construction or manufacturing in 2000. The latter delineation serves as a proxy for studying the effects of layoffs, as construction and manufacturing were the two most-affected industries in the Recession by a considerable margin.

Table 3 presents the results of this exercise. Two results quickly emerge: the long-run effects of the Recession on college attainment for middle-income children do not differ at all by whether their parents worked in heavily affected industries, but heterogeneity of effects over home ownership is quite sharp. Middle-class children whose parents did not have any house value do not observe any long-term effects of the Recession on college attainment that can be distinguished from zero, suggesting that the declines in their educational attainment can be almost *entirely* attributed to losses in parental home equity.

These results are consistent with a model wherein parent income and wealth (here captured by home equity) factor differently in financing higher education. Such a scenario may happen if middle-class parents consume their income in a hand-to-mouth manner but allow shocks to wealth to influence how much they finance the educational attainment of their child. This story is consistent with multiple entries of the economics literature: Hilger (2016) finds little evidence that parental layoffs between 2000 and 2009 depressed the college attainment of their children, but Lovenheim (2011) finds that house equity increases following the housing boom of the early 2000s substantially increased the college attendance of affected

Table 3: Investigation of Mechanisms for Middle-Class Children

VARIABLES	(1) College	(2) College	(3) College	(4) College	(5) College
Age 17 to 19	-0.022	-0.012	-0.009	-0.034	0.000
	(0.024)	(0.031)	(0.044)	(0.027)	(0.029)
Age 20 to 22	-0.354	-0.400	-0.126	-0.407	-0.293
	(0.101)	(0.108)	(0.189)	(0.137)	(0.115)
Age 23 to 25	-1.434	-1.504	-0.970	-1.047	-1.482
	(0.317)	(0.297)	(0.478)	(0.342)	(0.397)
Age 26 to 28	-2.373	-2.801	-0.838	-2.229	-2.317
	(0.371)	(0.404)	(0.547)	(0.506)	(0.466)
Sample	Baseline	HV > 0	HV $= 0$	C/M	Not C/M
Observations	303000	234000	69000	106000	196000
R^2	0.219	0.236	0.173	0.180	0.230

Notes: Standard errors clustered at the year-2000 CZ level and are in parentheses. Coefficients and sample counts rounded for disclosure avoidance purposes. The table displays estimates of β_a from equation (1) for the outcome variable of attainment of at least a four-year college degree, with age grouped into five bins and effects for the bottom age group normalized to zero. HV: parent house value in 2000. C/M: household head worked in construction or manufacturing in 2000. Not C/M: household head did not work in construction or manufacturing in 2000. The sample includes individuals of the 1990-1994 birth cohorts observed in the 2005-2018 ACS and linked to themselves in the 2000 Long-Form Census; see text for details.

families, with particular strong effects for families with relatively low household income[14].

4 Model

The empirical results described in Section 3 suggest that decreases to parental resources may have depressed the human capital accumulation of middle-class children in the Great Recession, with parental home equity being a potentially important mechanism. However, the analysis above is limited in several ways. The ACS sample allows me to observe individ-

[14]Indeed, using 2007-2009 HPI shocks from the Federal Housing Finance Agency as the source of variation instead of unemployment shocks results in a very similar pattern to the baseline results; see Figure A.6.

uals up through age 28 and therefore cannot speak to lifetime impacts of recession exposure. Additionally, the regression specification does not allow me to identify general equilibrium effects that affected all individuals, which means that the data are silent on the aggregate implications of the Great Recession on human capital accumulation. This "missing intercept" problem may be salient in this context, since individuals could be quite sensitive to macroeconomic as well as local conditions when making college enrollment decisions.

An economic model will allow me to study lifecycle and aggregate effects of the recession as well as assess the potential role of counterfactual policies in response to it. To this end, I develop a model that tracks the education decisions and life-cycle wages and earnings of adolescents. A period is one year. Children start the model at age 18 after having graduated high school and are endowed with a level of ability. The agent's parents start the model with a certain level of income and wealth, which may include both housing and non-housing assets. Parents provide an endogenous college transfer to their children who in turn select whether to pursue any postsecondary education. The agent faces a menu of college attendance options, and the likelihood of actually obtaining a college degree depends on both the agent's ability and the college option selected. After education outcomes are realized, the agent enters the labor force and makes labor supply and savings decisions every period until retirement at age 65.

This sequence of decisions closely follows those in Colas et al. (2021), with several key distinctions. First, agents have a menu of college options to choose from, allowing for potential substitution to cheaper but lower-quality schools in the face of parent resource shocks. Second, agents face both income and *employment* risk over the life cycle: choosing to participate in the labor market does not guarantee employment, which is critical to consider when evaluating the impacts of the Great Recession, as it featured a cratering of employment with fairly stable (or even slightly rising) real wages (Grigsby, 2023). The probability of employment will depend on educational attainment and accumulated experience, providing additional motivations for human capital investment in my framework. Third, wages are determined by accumulated experience instead of age as in Colas et al. (2021), allowing agents to endogenously adjust their labor force participation behavior in the face of economic contractions with the knowledge that doing so affects their future earnings trajectory.

The following sections lay out the decision problems solved by agents in the model in more detail before turning to the model's solution and estimation.

4.1 Decision Problems

4.1.1 Parent Transfer

Agents begin the model at age 18 with a continuous level of ability θ, the range of which is normalized to be 0 to 1, that influences college tastes and completion probabilities as well as earnings later in life. Parents start with income I^p and wealth W^p and choose a transfer t to give to their child to finance education. Importantly, W^p may include both housing and non-housing wealth. Specifically, the parent maximizes:

$$V^p(I^P, W^p, \theta) = \max_{t, t_H} \mathbb{E}[\log(I^p - t'_I) + \omega \log(W^p - t'_W) + \alpha V^c(\theta, t, I^p)]; \tag{2}$$

$$t = t_I + t_W. \tag{3}$$

The parent trades off utility from helping their child with their own consumption. The parent also chooses how to split the total transfer between transfers from their income and wealth. The child's enrollment decision and educational outcomes will be subject to idiosyncratic shocks that I assume parents are ignorant of before committing to a transfer schedule[15] — as a result, the parent gains utility from the expected utility of their child and their expected *realized* transfers t'_I, t'_W. Parents are altruistic and place weight α on the utility of their child, allowing for the possibility of crowd-out from policies that expand financial aid. The parameter ω captures how parents weigh utility from wealth versus utility from income, which will enable the model to reproduce different college enrollment elasticities following parental income and wealth shocks.

While parental transfers are endogenous in that they are sensitive to their child's future utility, I do not model the life-cycle labor supply and savings decisions of parents that transpire beforehand. Doing so would be highly complicated, as it would require modeling the accumulation of both housing and non-housing wealth as well as how such decisions respond to child characteristics. Moreover, such decisions are not germane to my research question, since the central experiment in the model will be to induce a shock that parents are unable to plan for ahead of time (i.e. the Great Recession). Considering the life-cycle decisions of parents is important if one is concerned about how parent savings behavior may respond to permanent changes to financial aid policies (Colas et al., 2021), but the unanticipated nature of the recession makes the parsimonious specification above preferable.

[15] This simplifies the analysis greatly through allowing these idiosyncratic shocks to be left out of the parent state space. Colas et al. (2021) adopt a similar structure.

4.1.2 Schooling Decision

Having received the transfer from their parents, the agent chooses whether to attend college. If the agent chooses not to go to college at all, they enter the working stage with zero savings. While college is typically modeled in the literature as a binary decision, the United States features a wide array of schooling options that differ considerably in their quality and cost. I allow for a broader menu of college-going options to reflect this and allow for potential substitution towards cheaper options in the face of parent resource shocks. I adopt a menu similar to the one used in Kennan (2021): agents may choose to not attend college at all or may attend a college of one of four types corresponding to community colleges, other public universities, and private colleges of two levels of quality. The agent chooses their preferred schooling option s, solving

$$V^c(\theta, \tau, I^p) = \max_s \left[\mathbb{E}[V_t^w(\theta, e, x, a)|s, t, I^p] + \psi_0 + \psi_1\theta\mathbb{1}(s \neq 1) + \psi_2\theta\mathbb{1}(s \in HQ) + \zeta_s\right], \quad (4)$$

where ζ_s is a vector of utility shocks distributed type 1 extreme value with spread parameter σ_ζ[16]. This functional form assumption allows for straightforward attendance probability calculations and for the child's expected utility in the parent transfer stage of the model to be solved for closed-form. In addition to idiosyncratic shocks, agents face fixed psychic costs for college enrollment that depends on their enrollment decision and ability: ψ_0 captures an average psychic cost for college attendance, while ψ_1 and ψ_2 capture varying psychic costs by ability for attending college at all and attending high-quality schools in particular.

The agent's continuation value V^w is formed over their ability θ, completed education e, experience x (which will always start at zero), assets a, and age t at the point they enter the labor market. Potential educational outcomes include school/college dropout ($e = 1$), some college/associate degree ($e = 2$), or bachelor's degree ($e = 3$). The probability of an agent achieving a certain level of schooling depends on both their selected schooling option and underlying ability, with the probability of reaching a bachelor's degree being larger for higher-quality schooling options and higher-ability individuals.

Education outcomes also influence the assets agents hold and the age at which they enter the working stage of the model. If agents do not pursue schooling whatsoever or drop out of college, they enter the model at age 18 with assets $a = 0$. If the agent obtains a bachelor's

[16] Another option is to normalize $\sigma_\zeta = 1$ and allow there to be an additional coefficient in front of V^w. The distinction is immaterial in terms of the model's results, and the current setup more resembles the college choice problem's structure in Heckman et al. (1998).

degree, they enter the working stage of the model at age 22 with starting assets equal to

$$a = t - (C_s - G(I^p, \theta, s)),$$

where C_s denotes the cost of schooling option s and $G(I^p, \theta, s)$ represents grants awarded to the child, which varies over parent income, ability, and choice of school. If the agent obtains some college or an associate degree, I assume that they start working at age 20 with assets $a = \frac{t-(C_s-G(I^p,\theta,s))}{2}$. If the agent drops out of college, they start working with a high school degree at age 19 with assets $a = \frac{t-(C_s-G(I^p,\theta,s))}{4}$. Importantly, agents face an educational borrowing constraint $\bar{d}$ and may not choose any schooling option that *could* result in their exceeding this limit. Discounting of V^w is done back to the beginning of the model to account for different ages of completing school.

4.1.3 Post-Graduation Labor Supply and Saving

In each period following their schooling decision, the agent chooses whether to supply labor. In addition to the state variables described above, I also track as a new state variable past labor force participation status (henceforth p) after the first working period to account for potential frictions in moving in and out of the labor market. Denoting $\Omega = \{\theta, e, x, a, p\}$ an agent who has made their labor supply decision $\ell \in \{0, 1\}$ and realized employment outcome (should they have participated) $m \in \{0, 1\}$ maximizes utility

$$V_t^w(\Omega; \ell, m) = \max_{c \geq 0} \left\{ \frac{c^{1-\eta}}{1-\eta} + \Lambda(\Omega, t)(1-\ell) + \beta \mathbb{E}[\tilde{V}_{t+1}^w(\Omega' | \Omega, \ell, m, c)] \right\} \tag{5}$$

subject to the budget constraint

$$c + a' = I(\Omega, m, \ell) + a(1 + r).$$

Consumption depends on how much the individual chooses to save for the upcoming period as well as their flow income $I(\Omega, m)$. If the individual does not participate in the labor market or is unemployed, the government provides transfers so as to guarantee a consumption floor $\underline{c}$. If the individual supplies labor and is employed, their flow income is $I(\Omega, 1) = w - \mathcal{T}(w)$, where w is their wage and $\mathcal{T}$ is the tax-transfer system. The wage is given by:

$$\log(w) = \gamma_0^e + \gamma_1^e \theta + \gamma_2^e x + \gamma_3^e x^2,$$

so wages depend on ability θ, work experience x, and experience, with separate values of γ for each possible education level (high school/dropout, some college/associate, bachelor's degree).

The second term of (5) captures the utility of leisure, which the agent only enjoys if they do not participate in the labor force (i.e. unemployed agents do not consume leisure). Utility from leisure is given by

$$\Lambda(\Omega, t) = \lambda_0 + \lambda_1\theta + \lambda_2 t + \lambda_3 t\mathbb{1}(t > 50),$$

so leisure utility is allowed to vary by agent ability and age. The final term of (5) is the continuation value: next-period utility is discounted by β, and the state space Ω evolves according to the choices made in the current period. Ability θ and education e are static, while experience x increments by 1 if the agent supplies labor and is employed. Assets a evolve according to $a' = \ell \cdot I(\Omega, m) + a(1+r) - c$, with the restrictions that individuals may not hold debt greater than the consumption floor $\underline{c}$ unless they do so from student loans, in which case they must, at minimum, make payments on student loans so that the debt is erased within 10 years of graduation. The previous-period participation status evolves according to $p' = 0$ if the agent does not participate, $p = 1$ if the agent is unemployed, and $p = 2$ if the agent is employed.

At the start of a period, the agent decides whether to supply labor, solving:

$$\tilde{V}_t^w(\Omega) = \max_{\ell}\left\{\ell \cdot \left[\sum_{m=0}^{1} \boldsymbol{Pr}(m|\Omega) \cdot V_t^w(\Omega; 1, m)\right] + (1-\ell)\cdot V_t^w(\Omega; 0, 0) + \xi S(\Omega, \ell) + \varepsilon_\ell\right\},$$

where ξ is a switching cost the agent must bear if they change their labor force participation status (i.e. $p \in \{1, 2\}$ and $\ell = 0$, or $p = 0$ and $\ell = 1$) after their initial working period, and ε_ℓ is a transient preference shock with mean 0 and standard deviation σ_ℓ. Finally, the probability of employment is given by

$$\boldsymbol{Pr}(m = 1|\Omega) = \Phi\Big(\pi_0 + \pi_1\mathbb{1}(p=2) + \pi_2\mathbb{1}(e=2) + \pi_3\mathbb{1}(e=3) + \pi_4 x + \pi_5 x\mathbb{1}(e=2) + \pi_6 x\mathbb{1}(e=3)\Big),$$

so the probability of employment varies based on experience, educational attainment, and previous employment. Finally, the agent retires at age 65, so $\tilde{V}_{66}^w(\Omega) = 0$.

4.2 Model Solution and Discussion

The model presented attempts to include several parsimoniously specified decisions surrounding early-life human capital investments and subsequent labor market outcomes that, when combined, allow for a rich set of behavioral responses to counterfactual scenarios and policies. If a recessionary shock is introduced to the model that worsens employment prospects, agents can attempt to supply more labor later in the life cycle to compensate for lost experience earlier on. At the same time, frictions to switching labor force participation status and worse employment prospects for individuals with less experience may result in economic contractions having long-lasting impacts on those who begin working during them. Parents can increase their transfers to their children to provide intergenerational insurance for a recession, though their ability to do so may also be affected if the shock impacts parental resources as well. Even if parental transfers are reduced, agents can elect to attend cheaper schooling options in an attempt to delay labor force entry and build more human capital. Thus, agents can employ a wide set of responses to various shocks in the model, and which options they choose may vary systematically over the parental income gradient.

Solutions to the agent's problems are computed via dynamic programming. The model is solved via backward induction, where policy functions and value functions are computed first for age 65, then age 64, and so on. When evaluating the objective function, I discretize the ability level into ten points and approximate continuation values via linear interpolation over the asset state. Policy functions for savings are computed for Brent's method for univariate optimization over a bounded interval. Expectations of $\tilde{V}_t^w(\Omega)$ over the ε_ℓ shock can be solved for closed-form by determining threshold levels of ε_ℓ for any state in the state space that govern labor force participation and applying the inverse Mill's ratio. After solving value functions for the working phase, closed form probabilities of school selections and child expected utility can be formed given a parent transfer t and child starting ability θ, allowing parent transfer policy functions to be computed as well.

5 Estimation

Estimation of the model proceeds in two steps: first, a subset of model parameters is calibrated outside of the model. The remainder of the model's parameters are estimated via the simulated method of moments. All monetary units in the model are scaled by 40,000 2012 dollars.

5.1 Sources of Estimation Targets

The main source of data I use to estimate the model is the restricted-use NLSY97 geocode, a dataset that surveys 8,984 youths aged 12-16 as of December 31, 1996. The NLSY97 is divided into two subsamples: a nationally representative sample of 6,748 youths and an oversample of 2,236 minorities. A crucial feature of the NLSY97 is that it contains both measures of ability (an individual's ASVAB/AFTQT score) and parental income and wealth. An additional important feature of the NLSY97 geocode is that it contains information on the exact institution a respondent is studying at should they choose to pursue any post-secondary schooling.

My estimation sample uses individuals who completed a high school degree by age 20 in the data and have non-missing ASVAB scores, parental income, and parental wealth[17]. Late college-going and return college-going are salient features of these data but are not modeled explicitly in my framework Kennan (2021) — to account for these, I code the first institution an individual is observed to be enrolled at up to age 25 as their choice of schooling and their educational attainment at up to age 29 as their final completed schooling. Individuals for whom final educational attainment is missing are dropped, resulting in a sample of 3,959 individuals. Longitudinal sample weights are used when computing all moments to target.

To assign ability levels to individuals in the NLSY97, I follow Dillon and Smith (2017) and use results from the ASVAB, a test designed for applicants to the U.S. military that most NLSY97 respondents took in 1997. The ASVAB has twelve separate component scores — I convert these scores to standard deviations within category and birth cohort (to account for individuals taking the test at different ages) before using the first principal component of the scores as my measure of ability. Consistent with Dillon and Smith (2017), I find that the first component explains roughly 60% of the total variance across the 12 sections and is strongly correlated with other test scores such as the ACT. I then compute percentiles of this measure and divide them by 100 to assign ability levels between 0 and 1 to individuals in the sample.

Since the NLSY97 sample is relatively small and only contains individuals born between 1980 and 1984, it does not contain adequate information about lifetime earnings profiles. To obtain lifetime profiles of earnings and labor force participation, I use data from the

[17]Measured as the sum of parental house equity and wealth held in prepaid tuition accounts, checking accounts, bonds and stocks, vehicles, real estate, businesses, and other, minus any money owed on vehicles, child payments, or other. Wealth in retirement accounts and pensions is not included as regulation regarding taking loans out of such accounts makes using them to finance college highly difficult.

2000 U.S. Census. I limit my sample to household heads aged 19 to 65 and restrict the sample in a manner similar to Huggett et al. (2011) when computing wage profiles to target in estimation. For household heads aged 30 or above, I require that individuals worked at least 520 hours in the previous year and made at least $1,500 1968 dollars in wage earnings. For household heads aged younger than 30, I require at least 260 hours worked and $1,000 1968 dollars. From these data, I then compute average earnings and the variance in log earnings by educational status and age as targets for my model to match. From the U.S. Census, I also obtain labor force participation and unemployment rates for household heads not enrolled in any higher education, again broken up by educational attainment and age, as additional targets in estimation.

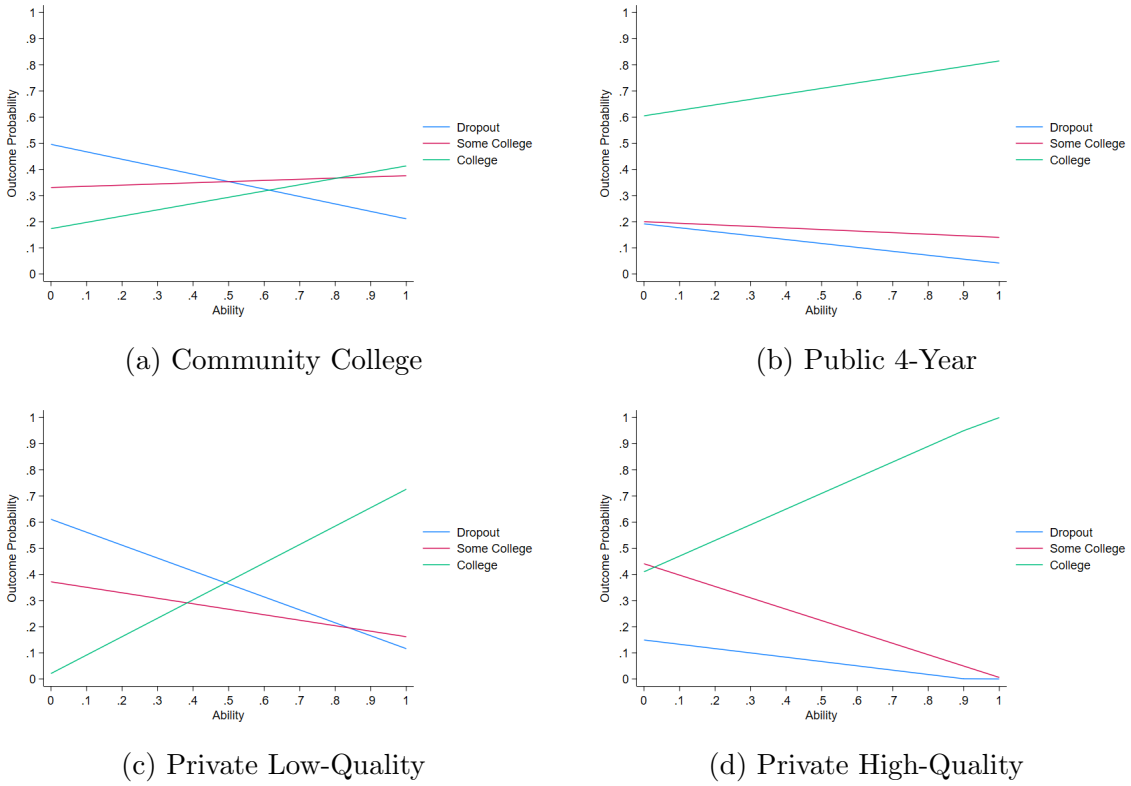
5.2 College Costs and Transitions

I use data from the Integrated Postsecondary Education Data System (IPEDS) to sort higher education institutions into the categories used in the model, which then allows me to calibrate the costs of schooling options. Additionally, since exact institutions of attendance are available in the NLSY geocode, these data can then be merged to the NLSY97 sample to compute moments of school attendance decisions by ability and parental resources as well as calibrate probabilities of educational outcomes conditional on agent ability and schooling selection.

In the model, agents may choose to attend community colleges, public four-year colleges, or private colleges of two levels of quality. College options differ in their cost, and probability of graduation, which also depends on the agent's ability. To obtain costs for each option, I use data from the 2000-2004 rounds of IPEDS data. I categorize private as high quality if they are both non-for-profit 4-year institutions and either 1) have a 25th percentile SAT math score of admitted students above 380, 2) have a 25th percentile ACT math score of admitted students above 16, or 3) have an admittance rate of less than 50%[18]. All other private institutions are coded as low-quality. The full cost of each schooling option is taken as the full-time-enrollment-weighted average sticker price (including tuition, room and board, and required fees) for schools in each category, multiplied by four. Costs for public institutions differ substantially based on whether students are attending in-state or out-of-state; when computing the costs of community colleges and other public universities, I weight the costs according to the proportion of individuals in the NLSY97 who attend such institutions in-

[18] Ishimaru (2023) categorizes institutions as selective or not using similar criteria.

Figure 6: Probabilities of Educational Outcomes over Ability by Enrollment Decision



(a) Community College

(b) Public 4-Year

(c) Private Low-Quality

(d) Private High-Quality

Notes: Figures present probabilities of educational outcomes based on ability level and enrollment decision. Data from IPEDS and NLSY97 geocode. See text for details.

state vs. out-of-state. This procedure results in calibrated schooling option 4-year costs of 0.84, 1.52, 2.56 and 3.41 ($33,600, $60,800, $102,400, and $136,560 in 2012 dollars).

Next, I merge the IPEDS data to data on enrollment choices and educational outcomes in the NLSY97 geocode. I calibrate the probabilities of dropout, obtaining some college, or obtaining a college degree by fitting a linear function of the outcome probability over ability terciles conditional on enrollment decision, and I impose that non-attendance results in the individual having a high school degree with certainty[19]. The transitions obtained from this process can be viewed in Figure 6. The probabilities of favorable educational outcomes conditional on college attendance choice are increasing in ability in all cases. Notably, low-ability children have a lower chance of obtaining a college degree at a high-quality private

[19]A very small proportion of individuals in the NLSY97 report having some college or a college degree despite never seeming to enroll at a postsecondary institution.

Table 4: Parameters Estimated Externally

Parameter		Value	Source
Tax schedule	$\mathcal{T}(w)$	Various	Heathcote et al. (2017)
Grant schedule	$G(I^p, \theta, s)$	Various	NPSAS
CRRA coefficient	η	2	Standard
Interest rate	r	0.03	Standard
Discount rate	β	0.97	Standard
Student loan limit	$\bar{d}$	0.735	Stafford limit
Consumption floor	$\underline{c}$	0.11	2000 Census

Notes: Table presents model parameters that are estimated externally. The first two columns describe the parameter and give its symbolic representation. Column 3 reports the parameter value when possible, and Column 4 reports the source of the value. Monetary units are scaled by 40,000 2012 dollars.

college than a four-year public university, while the reverse is true for high-ability individuals.

5.3 Parameters Calibrated Outside the Model

A summary of additional parameters calibrated outside the model may be found in Table 4. Taxes $\mathcal{T}(w)$ are calculated using the approximation in Heathcote et al. (2017), which has been shown to do well in representing U.S. tax policies. Grant schedules $G(I^p, \theta, s)$ are taken from the 2000 National Postsecondary Student Aid Study (NPSAS)[20], a nationally representative survey of 50,000 college students that contains detailed breakdowns of types of grant aid received by parent income, high school GPA, and type of institution attended. Utility over consumption is assumed to be CRRA with relative risk aversion parameter $\eta = 2$, which is roughly in the middle of values used in the empirical human capital theory literature. I set the risk-free interest rate to be $r = 0.03$ and assume $\beta = 1/(1 + r)$.

The borrowing limit $\bar{d}$ is set to \$29,720, which corresponds to the nominal Stafford borrowing limit of \$23,000 in 2000, which is the same limit used in Colas et al. (2021). The consumption floor $\underline{c}$ is set to roughly \$4,500, which comes from average income from welfare, social security, supplemental security, and "other" sources among unemployed individuals in the 2000 Census who provided 0 hours of work in the previous year and is quite similar to the value used in Fan et al. (2024).

[20] See https://nces.ed.gov/surveys/npsas/.

5.4 Method of Moments

After the calibration discussed above, I am left with 31 parameters to estimate via the simulated method of moments. These parameters are collected as

$$\Theta = [\omega, \alpha, \psi_0, \psi_1, \psi_2, \sigma_\zeta, \gamma_0^e, \gamma_1^e, \gamma_2^e, \gamma_3^e, \lambda_0, \lambda_1, \lambda_2, \lambda_3, \xi, \varepsilon_\ell, \pi_0, \pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6].$$

The first two parameters $[\omega, \alpha]$ represent the utility weight parent's place on wealth as well as the altruism parameter for the utility of their child. One challenge is that because parent income and house equity are strongly correlated, the parameter ω that represents the parent's utility weight on wealth is somewhat difficult to identify credibly. I overcome this by fitting a moment reported by Lovenheim (2011), who leverages plausibly exogenous housing wealth shocks to study the impact of family wealth on college attendance. He finds that an additional \$10,000 in parent house equity increases the probability of college attendance by the child by 0.7 pp. This provides a credible target for identifying ω via indirect inference, as I can easily simulate the probability that an individual attends college in a counterfactual world where their parents have an additional \$10,000 in housing wealth and compare this probability to the baseline simulation. I additionally target rates of college attendance by parent income and wealth equity quartile to target α as well as an elasticity reported by Hilger (2016) that a decline in parent income of approximately \$8,000 decreases the college enrollment of their children by approximately 0.43 percentage points.

The set of parameters $[\psi_0, \psi_1, \psi_2, \sigma_\zeta]$ correspond to psychic costs associated with college attendance as well as the spread of college preference shocks. These parameters are identified by targeting a rich set of college enrollment profiles from the NLSY97. For each ability tercile and parent income/wealth quartile, I compute the share of children who select each educational option (non-enrollment, public/private low/high quality) and use these moments to discipline the parameters. The psychic costs are identified by targeting overall college attendance rates as well as differences in enrollment and high/low quality selections by ability while conditioning on parent resources. Variation in selections made *within* ability/parent resource combinations identifies σ_ζ.

The next set of parameters $[\gamma_0^e, \gamma_1^e, \gamma_2^e, \gamma_3^e]$ govern earnings over the life cycle. These are identified by targeting life-cycle wage profiles by educational status. The effects of ability on earnings γ_1^e, conditional on education, are identified by matching the early-life ratios (as opposed to levels, since the sample composition of the NLSY97 considerably from the 2000

Census, from which other income moments are drawn) of wages of individuals in the top and bottom ability terciles within educational attainment categories in the NLSY97.

The parameters $[\lambda_0, \lambda_1, \lambda_2, \lambda_3, \xi, \varepsilon_\ell]$ control labor force participation in the model. I discipline these parameters to match labor force participation rates over the life cycle and by educational attainment. Uncertainty of labor force participation within a given age and educational attainment is useful in identifying the spread of LFP preference shocks ε_ℓ. Finally, to identify the LFP switching cost ξ, I construct a panel sample of labor force participation and employment over the life cycle from the Panel Study of Income Dynamics (PSID), following a data cleaning procedure that closely follows my treatment of the Census data in Section 5.1. Using the PSID sample, I compute the rate at which individuals either supply labor when they did not do so in the previous period, or vice-versa. This rate of change for labor force participation status from year to year is useful in disciplining ξ.

Finally, the parameters $[\pi_0, \pi_1, \pi_2, \pi_3, \pi_4, \pi_5, \pi_6]$ govern the probability of employment for individuals who choose to supply labor. These parameters are targeted to match employment rates over the life cycle by educational attainment. Finally, from the PSID sample described before I compute the probability of being unemployed conditional on supplying labor and either being employed or not employed in the previous period. To again account for the PSID having a different sample composition than the 2000 Census, I take the ratio of these two probabilities to create a moment that is highly informative in identifying π_1.

I simulate the college attendance and life-cycle earnings for 50,000 individuals. When drawing parental resources for each individual, I separate parent income into ventiles and estimate the distribution of child ability conditional on parent income ventile as a truncated normal distribution in the [0, 1] interval. I then estimate the probability that the parents own a house (as opposed to rent) for each parent income ventile in the NLSY97: conditional on owning a house, I assume that parental house equity follows a log-normal distribution specific to each income ventile. Non-housing wealth is parameterized similarly and is drawn independently of house equity[21]. This allows me to draw parental homeownership and house equity realizations for the child, after which I may compute parental transfers, draw schooling utility shocks and education realizations, and simulate subsequent life-cycle earnings while approximating the savings policy function via linear interpolation.

[21] While positive, the correlation between house equity and non-housing wealth is fairly weak in the data. The distributions are computed for parents with equity/non-equity wealth falling between the 25th and 75th percentile (within each income decile) to prevent wealth outliers from raising the spread of the distributions to unreasonable levels.

Table 5: Parameters Estimated via SMM

Parameter		$\hat{\theta}$	$\hat{\sigma}$	Targeted Moment
Parent Preferences				
Wealth weight	ω	1.189	1.626	Lovenheim (2011) elasticities
Altruism	α	0.094	0.008	Hilger (2016) elasticity
College Preferences				
Fixed cost	ψ_0	-24.00	0.799	Overall enrollment rate
Ability effect	ψ_1	57.89	1.209	Enrollment profiles by ability
Ability-HQ effect	ψ_2	15.67	0.082	Enrollment profiles by ability
Spread of utility shock	σ_ζ	13.89	0.840	Within-ability enrollment profiles
Earnings Determinants				
Constant, HS	γ_0^1	-0.376	0.178	Lifecycle earnings profiles
Ability effect, HS	γ_1^1	0.353	0.254	Lifecycle earnings profiles
Experience (linear), HS	γ_2^1	0.047	0.010	Lifecycle earnings profiles
Experience (quadratic), HS	γ_3^1	-0.001	5e-7	Lifecycle earnings profiles
Constant, SC	γ_0^2	-0.291	0.171	Lifecycle earnings profiles
Ability effect, SC	γ_1^2	0.246	0.250	Lifecycle earnings profiles
Experience (linear), SC	γ_2^2	0.053	0.008	Lifecycle earnings profiles
Experience (quadratic), SC	γ_3^2	-0.001	6e-7	Lifecycle earnings profiles
Constant, Coll	γ_0^3	-0.447	0.176	Lifecycle earnings profiles
Ability effect, Coll	γ_1^3	0.400	0.225	Lifecycle earnings profiles
Experience (linear), Coll	γ_2^3	0.062	0.009	Lifecycle earnings profiles
Experience (quadratic), Coll	γ_3^3	-0.001	1e-6	Lifecycle earnings profiles
Leisure Preferences				
Constant	λ_0	3.488	0.298	Overall LFP rate
Ability effect	λ_1	-1.158	0.456	LFP by education level
Age effect	λ_2	-0.081	0.009	Lifecycle LFP profiles
Age effect ($t > 50$)	λ_3	0.369	0.061	Lifecycle LFP profiles
LFP switch cost	ξ	12.32	0.232	LFP switching rate
Spread of LFP shock	σ_ℓ	7.560	0.222	Lifecycle LFP profiles
Employment Determinants				
Constant	π_0	0.996	0.147	Overall employment rate
Previously employed	π_1	0.738	0.099	Employment by previous status
Some college effect	π_2	0.305	0.167	Employment rate by education
College effect	π_3	0.480	0.163	Employment rate by education
Experience effect	π_4	0.003	0.007	Lifecycle employment profiles
Experience x Some college	π_5	-0.006	0.009	Lifecycle employment profiles
Experience x College	π_6	-0.009	0.009	Lifecycle employment profiles

Notes: Table presents model parameters estimated via indirect inference. Columns 1 and 2 describe parameters and present symbolic representation. Columns 3 and 4 present parameter estimates and standard errors, obtained via Gourieroux et al. (1993). Column 5 describes data moments used to discipline parameters. See text for details.

Denoting $M = [M_1, M_2, ..., M_N]$ as the vector of empirical moments I target in the simulation procedure, denote $g(\Theta)$ as the vector of percentage errors between the data moments and the simulated moments, so:

$$g(\Theta)_i = \left(\frac{M(\Theta)_i - M_i}{M_i} \right), \quad i = 1, ..., N,$$

where $M(\Theta)_i$ is the ith moment simulated from the data given the parameter guess Θ. I find the point estimate $\hat{\Theta}$ by numerically solving

$$\hat{\Theta} = \arg\min g(\Theta)' W g(\Theta), \tag{6}$$

where $M(\Theta)$ are the simulated model moments and W is the identity matrix, with the exception that specific moments that are especially helpful for identifying specific parameters are given more weight. Minimization of the objective function proceeds via the Sbplx[22] routine. I compute standard errors by evaluating the numerical gradient of the objective function and applying the standard indirect inference formula (Gourieroux et al., 1993).

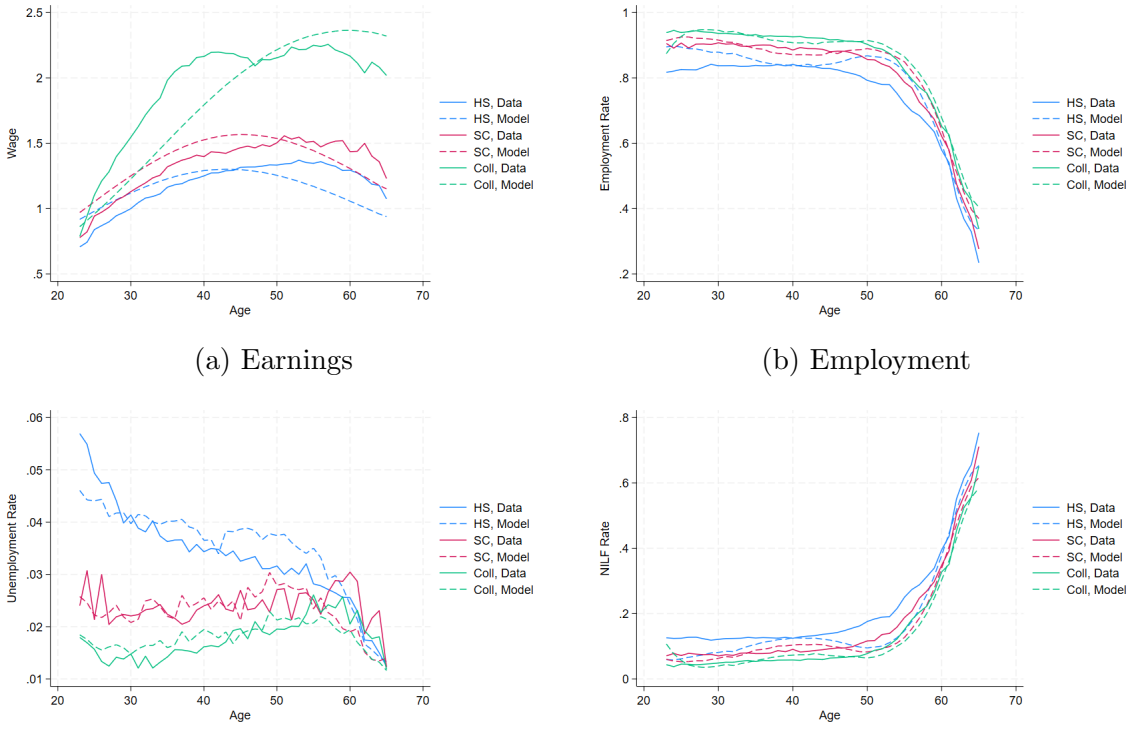
5.5 Goodness of Fit

The values of the parameters estimated via SMM are reported in Table 5. The weight on wealth in the parent utility function is greater than one, which will result in parents being less willing to draw from savings than flow income when providing transfers to their children. As a result, shocks to income and wealth can have different implications for college attendance propensities, particularly given that need-base grants are sensitive to the former. The preferences of the child over college attendance results in parental transfers having a large impact on child utility, especially if they move the child under borrowing constraints for particular schooling options — hence, even a low level of parental altruism can result in generous parent behavior.

The psychic costs associated with college and the scale of the college preference shocks are large, which is a similar result observed in other entries of the literature (Heckman et al., 1998), and preferences for college attendance (especially high-quality college attendance) are strongly increasing in ability. The wage estimates feature comparable intercepts across educational levels but different returns to experience, allowing for the wages of college graduates

[22] A variant of the Subplex routine, which itself executes Nelder-Mead on a sequence of subspaces. See https://nlopt.readthedocs.io/en/latest/NLopt_Algorithm/.

Figure 7: Model Fit — Earnings and Labor Force Profiles



(a) Earnings

(b) Employment

(c) Unemployment

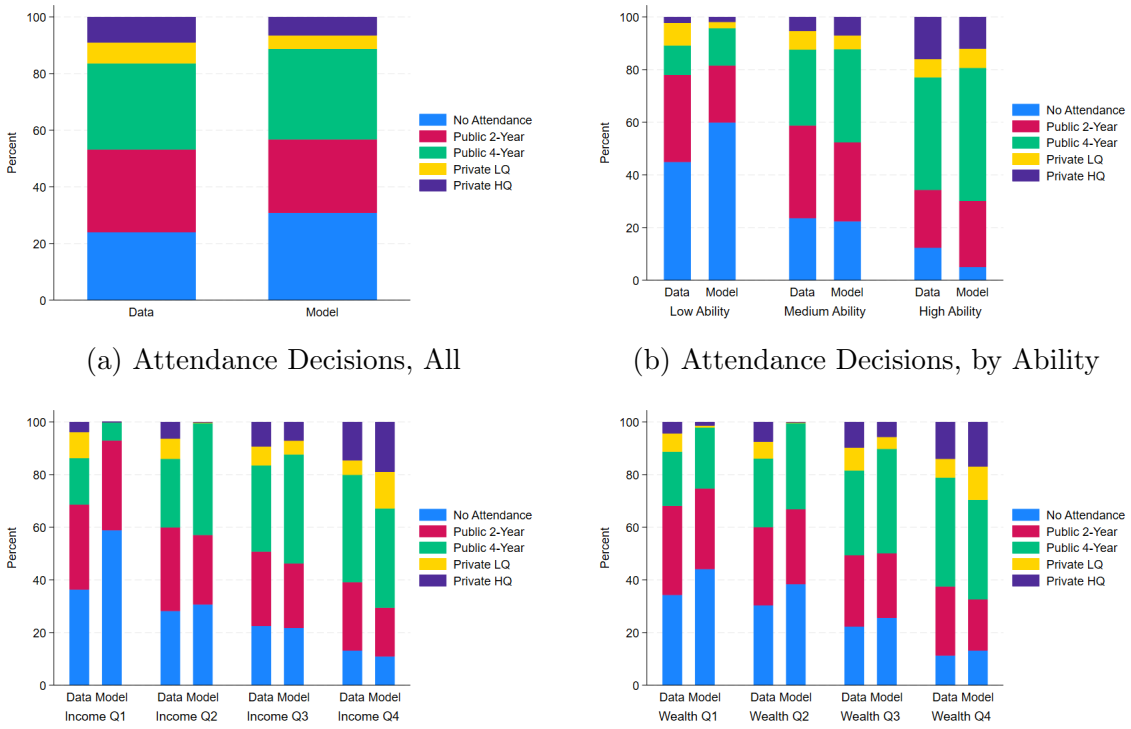
(d) Labor Force Non-Participation

Notes: Figure presents age profiles of wages and employment outcomes from data and model simulation. Moments from 2000 Census. Earnings normalized by $40,000. See text for further details.

to grow faster over the life cycle despite starting off similar to the wages of non-college graduates. Notably, the wage returns to ability are highest for college graduates and high school graduates and lowest for individuals who exactly obtain some college, indicating that the latter category may be especially likely to put individuals on relatively low-variance wage paths.

Utility for leisure is negative in ability, which is needed to match lower rates of non-participation of college graduates even with their higher wages. Switching costs and the spread of LFP shocks are both large, suggesting considerable non-pecuniary factors going into labor force participation decisions. Being previously employed and having either some college or a college degree are all associated with substantially improved employment prospects, and the returns to experience in employability suggest a role for experience that, while positive,

Figure 8: Model Fit — College Attendance Decisions



(a) Attendance Decisions, All

(b) Attendance Decisions, by Ability

(c) Attendance Decisions, by Parent Income

(d) Attendance Decisions, by Parent Wealth

Notes: Figure presents shares of college attendance decisions (no enrollment, public 2-year, public 4-year, or private low and high-quality) from data and model simulation. Moments from NLSY97 geocode and IPEDS. See text for details on sample construction.

eventually fades.

Figure 7 reports the fit of the model with regard to age profiles of earnings and employment. The model obtains a reasonable fit of wage profiles, though it understates the difference between college graduates and non-graduates in labor force non-participation. Additionally, high school graduates and those with more education have different unemployment profiles in that the former exhibits a clear decline over the life cycle, while the unemployment rates of individuals with some college or more hold fairly steady as they age.

The model's fit with regard to college attendance rates may be found in Figure 8. Across all attendance decisions, the model's fit is reasonable, though it does slightly overstate rates of non-attendance. Breaking the fit of college attendance down by ability level and parental resources reveals that the model especially overstates college non-attendance for the lowest-

Table 6: Model Fit — Other Moments

Moment	Data	Model	Source
Labor Market Frictions			
Labor force switching rate	0.032	0.037	PSID
Switch rate, entering*	0.19	0.16	PSID
Switch rate, leaving*	0.02	0.02	PSID
$P[\text{m=1}|p=2]/P[\text{m=1}|p\neq 2]$	0.21	0.23	PSID
College Enrollment Elasticities			
Effect of \$10k↑ house equity on enrollment	0.71	0.75	Lovenheim (2011)
House equity elasticity, income<\$70k	5.67	1.00	Lovenheim (2011)
Effect of layoff on enrollment	-0.43	-0.46	Hilger (2016)
Loans held at 25*	0.15	0.17	NLSY97
Ability Wage Effects			
High/low earnings ratio, High School	1.26	1.24	NLSY97
High/low earnings ratio, Some College	1.17	1.17	NLSY97
High/low earnings ratio, College	1.23	1.28	NLSY97

Notes: Table presents model fit of additional moments used in estimation. Moments from PSID, Lovenheim (2011), and Hilger (2016). A * indicates a moment not explicitly used when evaluating Equation 6. See text for sample construction.

ability and lowest-resource children, while the model's fit of college attendance profiles for middle to upper-class children is more successful.

Finally, Table 6 reports the model's fit of other moments used in estimation. The model generates reasonable levels of labor force participation switching and the relative effect of previous employment on current employment prospects. The model can also reproduce that most labor force switches are made by individuals entering as opposed to leaving the labor market despite these statistics not being explicitly targeted in the estimation procedure. The model also produces relative wages between high and low-ability individuals, conditional on schooling, compared to the NLSY97.

The model also is successful in fitting elasticities of college enrollment in response to parental resource shocks reported in other literature entries, which I can evaluate by directly computing the change in likelihood of college enrollment for individuals in the model under counterfactual parent resource scenarios. The model generates an effect of a \$10,000 increase in housing wealth on college enrollment similar to the effect reported in Lovenheim

(2011). Consistent with the paper, the model also predicts that these effects are localized for relatively poor children: an additional $10,000 in home equity increases the likelihood of college attendance for children with parental income less than $70,000 by 0.94 percentage points, while the corresponding statistic for children with parental income above $70,000 is 0.11 points. The effect for poor children is still lower than Lovenheim (2011)'s estimate of 5.67 points but lies well within the estimate's confidence interval. I also introduce parental layoffs in the model by reducing parental income by 13.74% and can match their impact on college enrollment pooled over the ages 18 to 22, as estimated by Hilger (2016). The model also generates sensible predictions of student loans held at age 25 compared to individuals in the NLSY97.

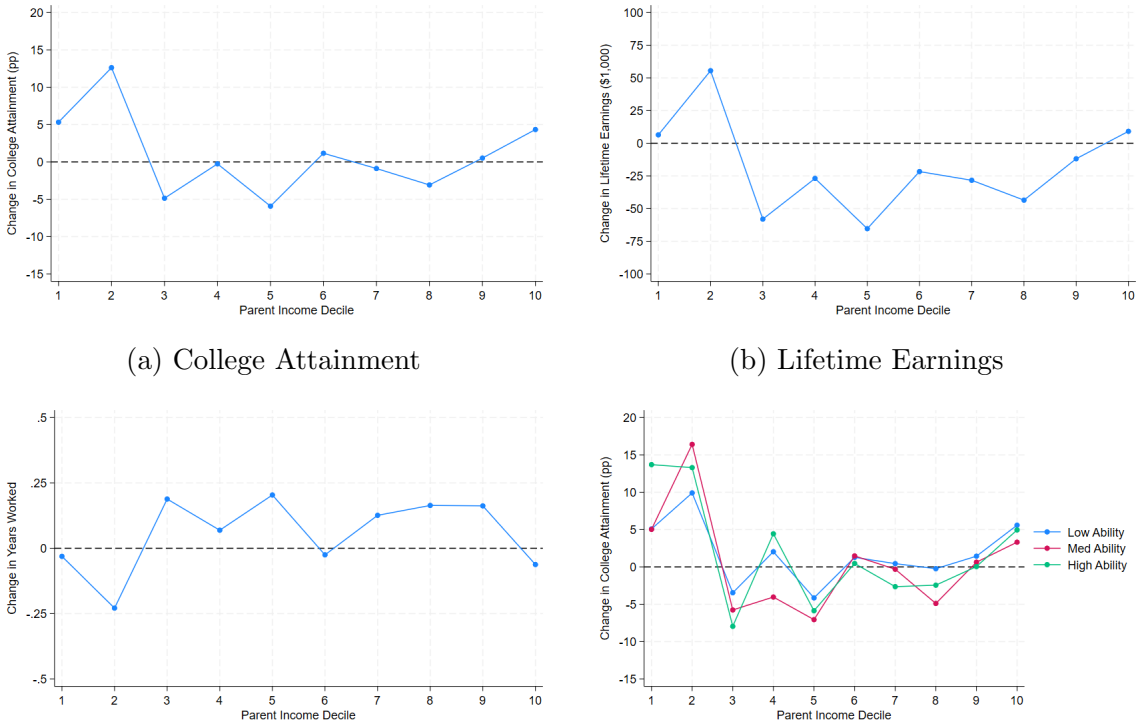
6 Counterfactual Scenarios

6.1 Impacts of a Recession

Having verified the model's ability to fit salient features of the data, I next turn to using the model to analyze the impacts of a recession. I simulate a contraction resembling the Great Recession in my model by introducing a five-year downturn with three main features. First, initial employment probabilities and employment transitions change for the worse. Agents may avoid some of this by enrolling in college and so forgoing the nadir of the recession, but college graduates will still face worsened prospects in the first year of their working life. Second, parental resources decline, and third, financial aid policy changes.

Modified employment probabilities come from the 2008-2013 waves of the American Community Survey. Specifically, I estimate employment probit coefficients for fresh graduates at each education level in the pre-recession (2008) wave of the ACS and then estimate the extent to which these coefficients changed for fresh graduates in subsequent years. I then modify values of π_0, π_2, and π_3 (the unemployment constant and education-specific effects) in the model by the same fraction as observed in the ACS time series for the first five years of the lifecycle. I calibrate the size of the housing wealth shock to -25.5%, which comes from population-weighted county-level real house price changes between 2007 and 2012 as measured in the Federal Housing Finance Agency House Price Index series. I also allow for potential parental layoffs by estimating, for each income ventile, the change in likelihood that adults in the NLSY79 experience an unemployment spell in the 2009-2013 period compared to the preceding four years and interpret this change as the additional layoff risk induced

Figure 9: Recession Lifetime Impacts



(a) College Attainment

(b) Lifetime Earnings

(c) Lifetime Labor Supply

(d) College Attainment, by Ability

Notes: Figures present effects of counterfactual recession on educational attainment, lifetime earnings, and lifetime labor supply. See text for details.

by the recession for parents of various income levels. I model layoffs as before as a 13.74% reduction in parental income.

Finally, I change financial aid policy by raising the Stafford Loan limit to $31,000 and by increasing the existing Pell grant schedule by roughly 25%, following the changes that were actually implemented in the wake of the financial crash (Barr and Turner, 2013). I also allow educational costs to rise in the wake of the recession, which I compute by observing how posted tuition, room and board costs changed across different types of universities in the 2008-2012 period, relative to trend, in the 2013 *Trends in College Pricing* report. After introducing these shocks, I re-simulate the outcomes of individuals in the new economy, allowing me to directly compare simulated individuals in the baseline model to their outcomes in the counterfactual world.

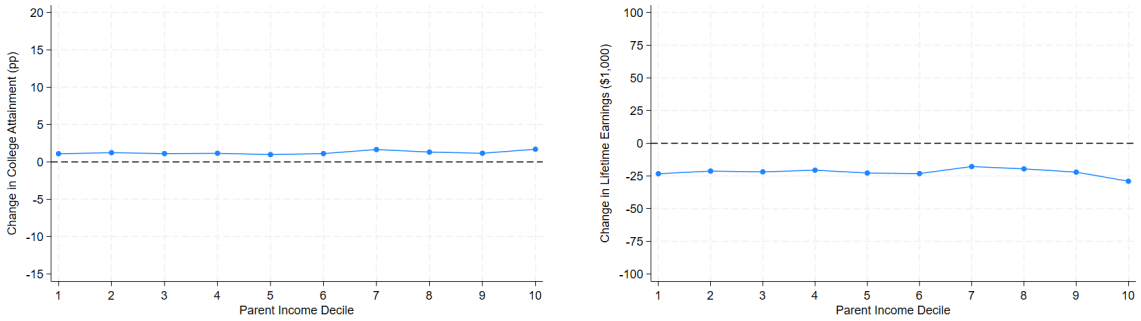
Figure 9 displays the results of this simulation. In the aggregate, college enrollment increases by roughly 2 percentage points in the recession, from 70.5 to 72.4 percent. This magnitude is consistent with the literature — Barr and Turner (2013) estimate that detrended college enrollment for 18-19 year old adolescents rose by a similar proportion in the 2007-2009 period. However, this aggregate measure masks considerable heterogeneity over the parent income gradient: Figure 9a demonstrates a heterogeneity very similar qualitatively to the results in the design-based analysis, with college attainment increasing for children in top and bottom deciles and decreasing for children in the middle of the income distribution. Low-income families frequently have low housing wealth or none at all, so parent housing wealth shocks have little impact on the direct financial cost a poor child has to bear in order to attend college. The same applies to children from high-income families: even with the house price shock, children from such rich families can obtain schooling while accruing minimal debt. The effects of the shock are especially positive for children in the second income decile, for whom the expansions in Pell grants and borrowing constraints were especially impactful in enabling them to stay under the borrowing constraint in attending four-year public universities. Decomposing the impacts further indicates relatively little heterogeneity over the child ability distribution (Figure 9d).

Accounting for intensive-margin adjustments in schooling decisions (i.e. attending lower-quality schools as opposed to not enrolling at all) is also important in arriving at this result. While the four-year college attainment of fifth-decile children declines by 6 pp, for instance, only half of this effect comes from these children directly choosing to forgo college all together. The other half of the effect comes from increased rates of college dropout and associate degree attainment following these children substituting into lower-quality schooling options. Thus, focusing only on a binary college enrollment decision could substantially underestimate the impacts of the recession on completed education.

Decreased educational attainment and worsened employment prospects result in the recession decreasing lifetime earnings for nearly all children (Figure 9b), though decreased educational attainment also means more time spent in the labor force, resulting in many children actually spending slightly more time working over the life cycle (Figure 9c). Like college attainment, though, losses in lifetime earnings are concentrated among middle-class children: fifth-decile children suffer losses in lifetime earnings of approximately $60,000, while children in the top and bottom parent income quartiles experience only a fraction of these losses or even gains in income due to increased college attainment.

To understand better the role that educational borrowing constraints had in generating

Figure 10: Recession Lifetime Impacts, No Borrowing Constraints



(a) College Attainment

(b) Lifetime Earnings

Notes: Figures present effects of counterfactual recession on educational attainment and lifetime earnings when student loan borrowing limits are not present in the model. See text for details.

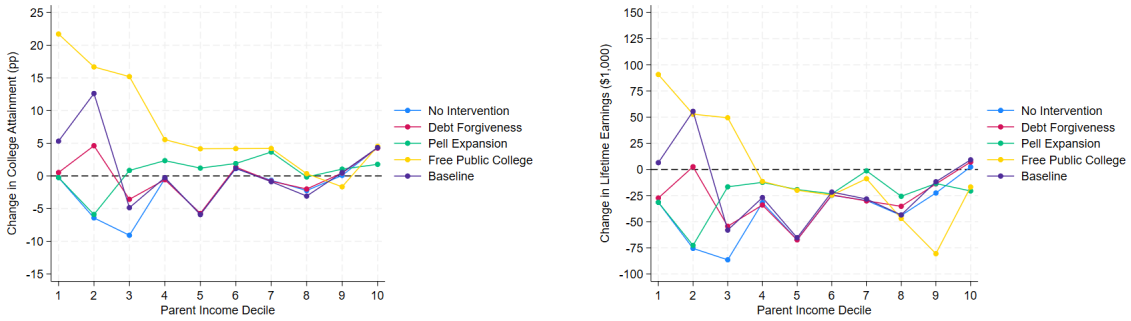
the "hollowing out of the middle" effect observed in the design-based and structural analysis, I next introduce an identical recessionary shock in a counterfactual setting where educational borrowing constraints are removed from the model entirely[23]. The results of this exercise are presented in Figure 10 and indicate that the hollowing phenomenon vanishes entirely when student loan limits are not present. While the increase in college attainment is similar in magnitude to the baseline model, this increase is spread evenly over the parent income gradient instead of being concentrated in the bottom and lower tails, which then translates to much more even losses in lifetime income. This result points to borrowing constraints playing a critical role in making middle-class children more vulnerable to losses in human capital following an economic downturn.

6.2 Counterfactual Interventions

To what extent could other government interventions mitigate losses of human capital during a recession? To probe this question, I conduct three policy counterfactuals designed to encourage college attendance and assess their impacts in and out of a recession. In the first experiment, I expand the borrowing limit by $10,000, and the first $10,000 of debt that

[23]The removal of borrowing constraints from the model increases college attainment by approximately 7.5 percentage points, which is considerably higher than the elasticity estimated by Johnson (2013). Taking more care in modeling student loan debt repayment plans and the burden they impose on consumption may enable the model to better reflect agent reluctance to take on debt in the first place.

Figure 11: Impacts of Recession with Different Policy Responses



(a) College Attainment

(b) Lifetime Earnings

Notes: Figures present effects of counterfactual recession on educational attainment and lifetime earnings when different policies are introduced in response to recession. See text for details.

college attendees take on is forgiven. In the third, the Pell benefit schedule is extended to provide the average Pell benefit to individuals with parent income up to $125,000. In the fourth experiment, community colleges and public universities are made entirely free for individuals with parent income up to $125,000. In each counterfactual, I simulate the college decisions and earnings of the same individuals during the recession to see how the policies might blunt the recession's impact if they are introduced during it.

I first study how these policies change the incidence of the recession if they roll out during the recession instead of the policies introduced in the baseline simulation. Figure 11 plots the effects of the recession on college attainment and lifetime earnings under the three counterfactual policies, the baseline simulation, and an alternate version where no policy change is introduced during the recession at all. Making college free for most agents in the midst of the recession results in college attainment for middle-class children and below rise dramatically, with children in the upper income deciles now sustaining the largest losses in lifetime earnings. Notably, the impacts of the debt forgiveness policy closely resemble the impacts of the baseline model, and the version of the simulation with no policy intervention tracks nearly identically to the expanded borrowing constraint policy for children in the third income decile or above. This indicates that the increased student loan limits and expansions to the Pell program were both important factors for encouraging college attendance for lower-class children in the financial crisis. The more generous Pell extension counterfactual succeeds in preventing human capital losses among the middle class, but since the policy

Table 7: Counterfactual Cost-Benefit Analysis

Policy	Baseline, No PBR	Baseline	Recession, No PBR	Recession
Debt forgiveness	-2.10	-2.99	-0.74	-2.89
Pell expansion	-2.88	-2.90	-3.69	-2.25
Free college	-28.03	-28.96	-32.29	-29.8
Recession policy	2.40	1.29	3.81	1.77

Notes: PBR = Parent behavioral response. The table reports changes in average per-capita NPV in thousands of dollars compared to baseline model. See text for details on simulation and NPV calculation.

does not alter borrowing constraints the educational attainment of the lowest-decile children is less responsive.

While counterfactual policies may be effective in mitigating the adverse effects of recessions, they may not be cost-effective for governments to implement. To evaluate this, I compute the net present value of government expenditure on children in the model as

$$NPV_i = \left(\sum_{t=1}^{T} \beta^t \mathcal{T}(w_{it}) \right) - G_i,$$

where i indexes individual model agents, t iterates over the life cycle, and G_i represents government expenditure on individual i, computed as the sum of Pell grants and forgiven debt, or the total cost of college attendance for qualifying individuals in the free-college counterfactual. For each policy, I re-simulate outcomes and calculate the average change in per-capita NPV in thousands of dollars compared to the baseline world.

These statistics are reported in Table 7, with separate entries for whether the policies are introduced in the recession version of the model and whether parents respond to the policies when making their transfer decisions. The more generous policies generally fail to be cost-effective regardless of whether they are implemented during an economic downturn and whether parents respond to them — this result is similar to those in Abbott et al. (2019) and Colas et al. (2021) in suggesting that progressive financial aid policies are more efficient.

The exception to this, however, is the policy combination that was introduced in the Recession in the real world (namely, the expansion to the Pell program and Stafford loan limits). The progressive nature of these policies lends to their cost-effectiveness — moreover, the simulation indicates that the policies are better investments when introduced in the recession state of the world compared to the baseline environment, regardless of whether parents respond to them when designing their transfers to their children.

7 Discussion and Conclusion

This paper studies how the effects of recessions on human capital and earnings vary over the parent income gradient. Using a novel confidential linked dataset from U.S. Census, I document that the countercyclicality of college enrollment and completion is much stronger for poor and rich children than it is for their middle-class counterparts. Other things held equal, exposure to a one percentage point larger 2007-2009 CZ unemployment shock in the recession is associated with a 7% long-run reduction in college attainment for children with parents in either the second or third quartile in the parental income distribution. The effects for children with bottom and top-quartile parents are a 1% gain or 2% reduction respectively, with neither effect being statistically significant. Higher rates of college non-completion appear to be more important in generating this finding than substitution to two-year degrees, and additional evidence points to losses in parental home equity being a particularly important mechanism for reducing college attainment among middle-class children.

The use of an economic model allows me to study the aggregate implications of these responses over the entire life cycle as well as to investigate how policy interventions might mitigate these impacts. The model replicates the patterns found in the data and documents a "hollowing out of the middle" wherein lifetime earnings gaps between middle-class and the wealthiest children rise considerably. Educational borrowing constraints are critical in generating this result, and while other policies may help mitigate these effects, achieving cost-effectiveness with them may be difficult due to parental crowd-out, and substitution into lower-quality college options. Nonetheless, governments may have a role to play in ensuring their long-run supply of human capital when the economy experiences a contraction, and progressive financial aid policy, along with reducing educational borrowing constraints, may be particularly worthwhile during downturns.

The main limitation of the empirical analysis is that outcomes are only observable up through age 28. Additionally, while much of the identification of the model comes through early-life earnings and wages, the model is limited in its ability to account for potential changes in late life-cycle labor force participation behavior among cohorts affected by the Great Recession. The model is also simplistic in how it models educational borrowing constraints and loan repayment. Considering the potential importance of private loans and how they vary over child ability (Lochner and Monge-Naranjo, 2011), along with more precisely modeling loan repayment schemes, would be highly desirable.

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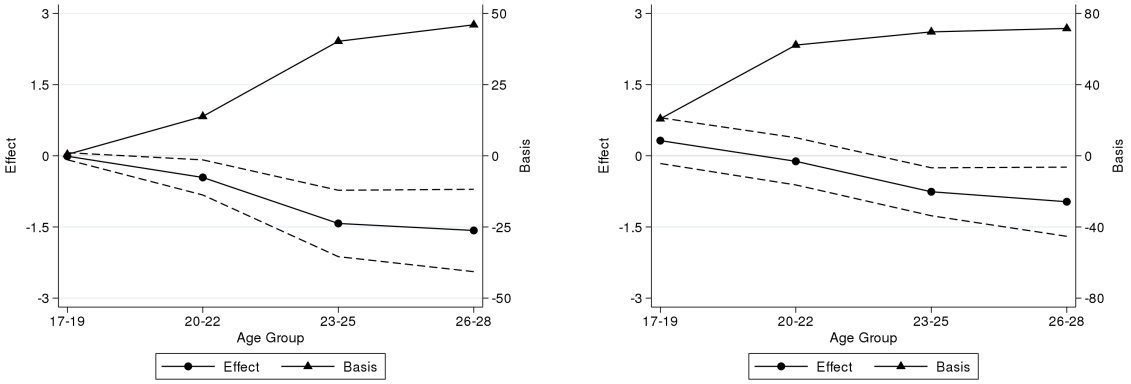
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A Supplementary Figures from Design-Based Analysis

Figure A.1: Full-Sample Effects of the Recession on other Educational Attainment Levels

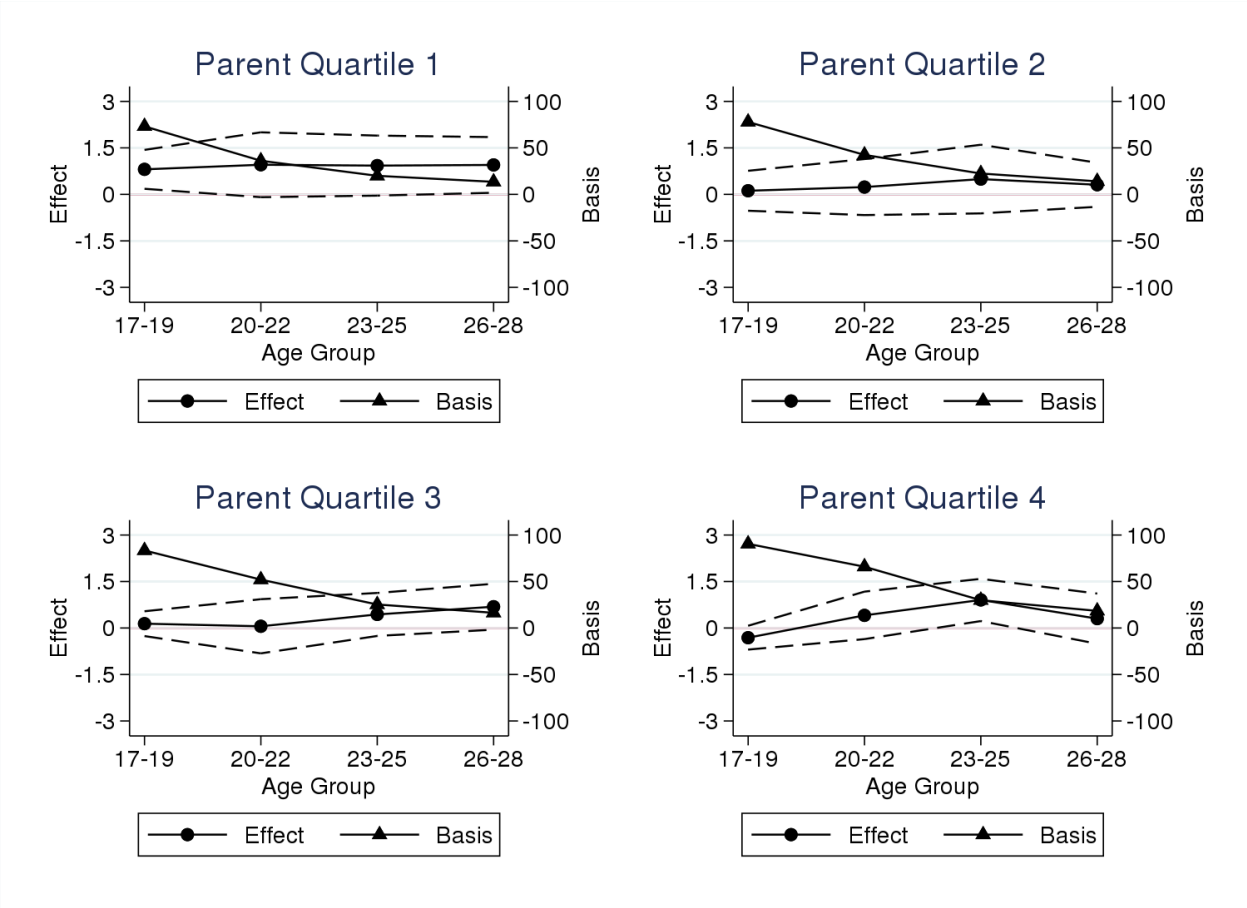


(a) Associate Degree

(b) Some College

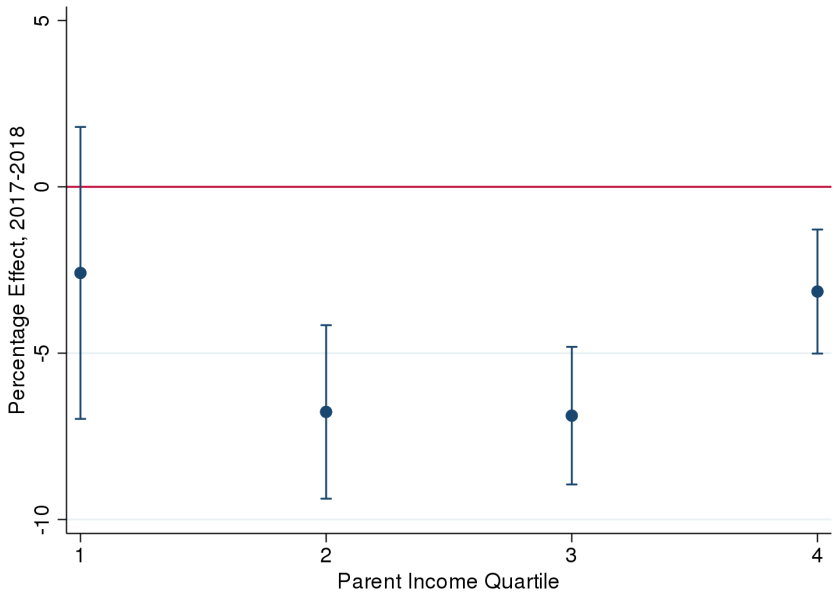
Notes: Figures display outcome variable means over age and estimates of β_a from equation (1), with age grouped into five bins and effects for the bottom age group normalized to zero. Outcome variables are dummies for attainment of at least either an associate degree or some college. Standard errors are clustered at the year-2000 CZ level, and the dotted lines indicate 95% confidence intervals for coefficient estimates. Coefficient estimates correspond to the circular dots and are measured by the left-hand axes; bases correspond to triangular dots and are measured by the right-hand axes. The sample includes individuals of the 1990-1994 birth cohorts observed in the 2005-2018 ACS and linked to themselves in the 2000 Long-Form Census; see text for details.

Figure A.2: Effects of Recession Shocks on School Enrollment by Parent Income Quartile



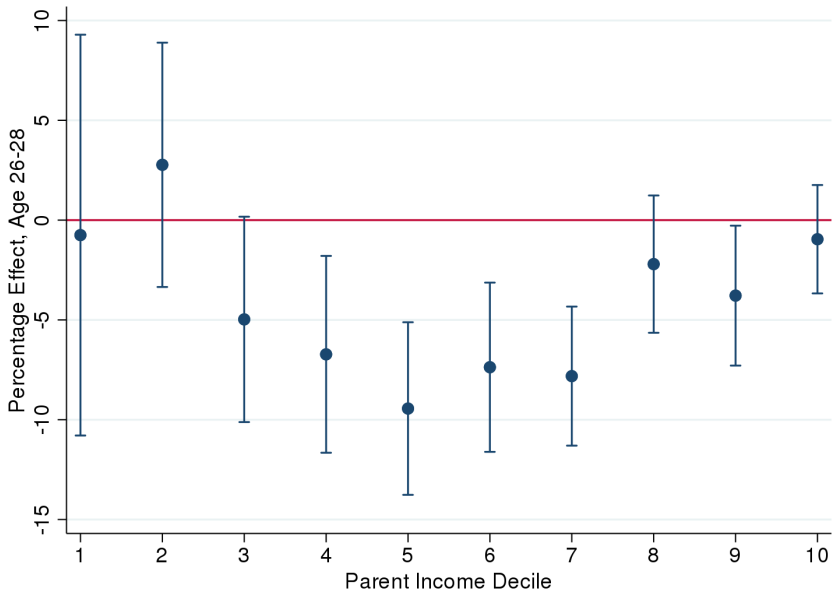
Notes: Figures display outcome variable means over age and estimates of β_a from equation (1) for the outcome variable of being enrolled in school at time of survey, with age grouped into five bins and effects for the bottom age group normalized to zero. Effects estimated separately over year-2000 parent income quartile. Standard errors are clustered at the year-2000 CZ level, and the dotted lines indicate 95% confidence intervals for coefficient estimates. Coefficient estimates correspond to the circular dots and are measured by the left-hand axes; bases correspond to triangular dots and are measured by the right-hand axes. The sample includes individuals of the 1990-1994 birth cohorts observed in the 2005-2018 ACS and linked to themselves in the 2000 Long-Form Census; see text for details.

Figure A.3: Percentage Effects in year 2017-2018 over Parent Income Quartile for College Attainment



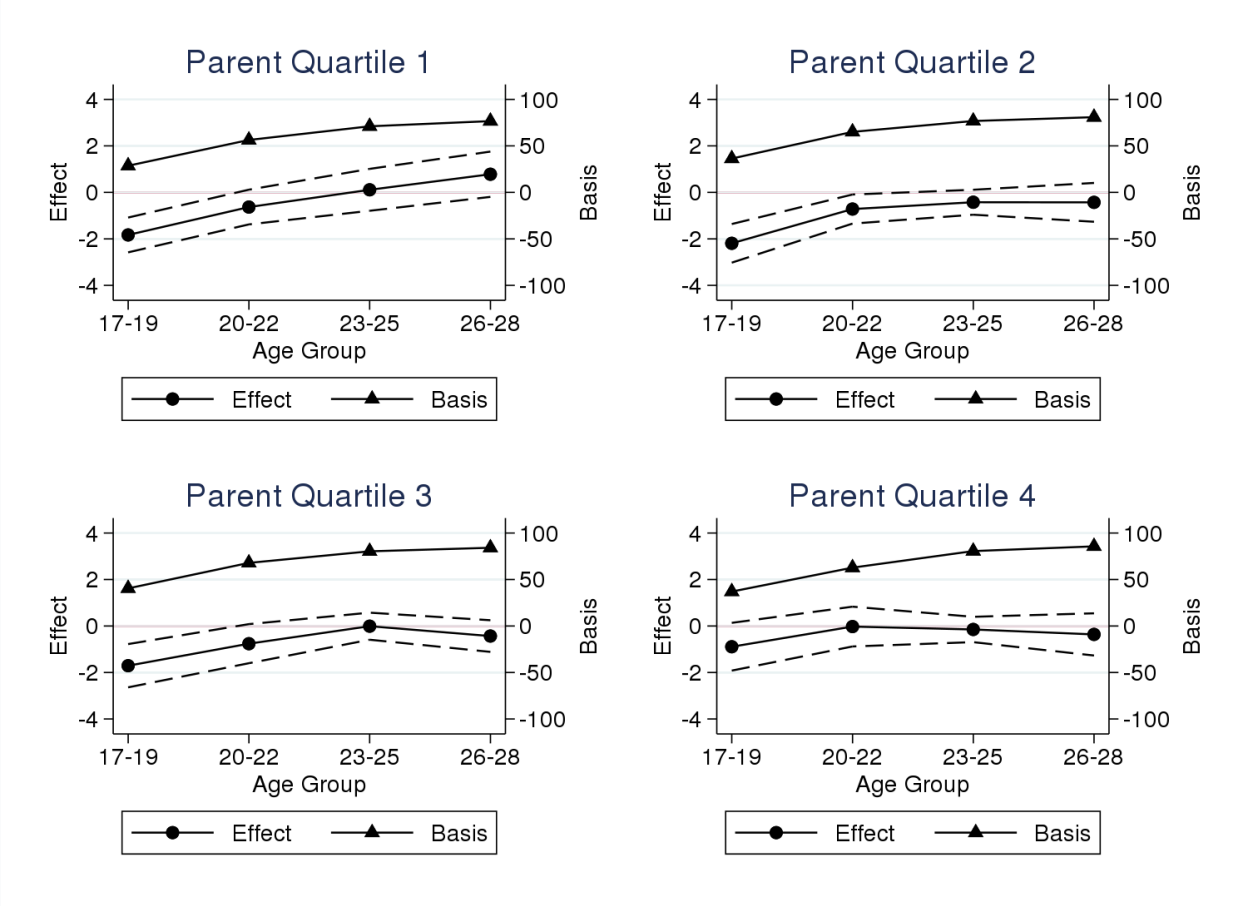
Notes: Figures display estimates of $\beta_{2017-2018}$ for the outcome variable of attainment of at least a four-year college degree over parent income decile. Coefficient and standard error divided by outcome variable mean for relevant parent decile group in year 2017-2018 to form percentage effects. Standard errors are clustered at the year-2000 CZ level, and 95% confidence intervals are displayed. The sample includes individuals of the 1990-1994 birth cohorts observed in the 2005-2018 ACS and linked to themselves in the 2000 Long-Form Census; see text for details.

Figure A.4: Percentage Effects at Age 26-28 over Parent Income Decile for College Attainment



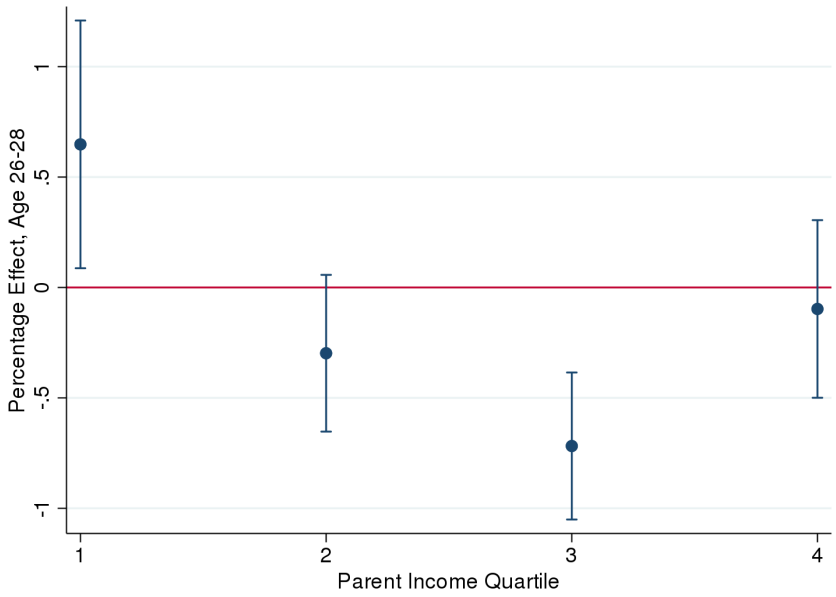
Notes: Figures display estimates of β_{26-28} from equation (1) for the outcome variable of attainment of at least a four-year college degree over parent income decile. Coefficient and standard error divided by outcome variable mean for relevant parent decile group at age 26-28 to form percentage effects. Standard errors are clustered at the year-2000 CZ level, and 95% confidence intervals are displayed. The sample includes individuals of the 1990-1994 birth cohorts observed in the 2005-2018 ACS and linked to themselves in the 2000 Long-Form Census; see text for details.

Figure A.5: Effects of Recession Shocks on Employment by Parent Income Quartile



Notes: Figures display outcome variable means over age and estimates of β_a from equation (1) for the outcome variable of being employed at time of survey, with age grouped into five bins and effects for the bottom age group normalized to zero. Effects estimated separately over year-2000 parent income quartile. Standard errors are clustered at the year-2000 CZ level, and the dotted lines indicate 95% confidence intervals for coefficient estimates. Coefficient estimates correspond to the circular dots and are measured by the left-hand axes; bases correspond to triangular dots and are measured by the right-hand axes. The sample includes individuals of the 1990-1994 birth cohorts observed in the 2005-2018 ACS and linked to themselves in the 2000 Long-Form Census; see text for details.

Figure A.6: Percentage Effects at age 26-28 over Parent Income Quartile for College Attainment using HPI Shocks



Notes: Figures display estimates of $\beta_{26-28}$from equation (1) for the outcome variable of attainment of at least a four-year college degree over parent income decile and CZ-level 2007-2009 HPI shocks used as variation instead of unemployment shocks. Coefficient and standard error divided by outcome variable mean for relevant parent decile group at age 26-28 to form percentage effects. Standard errors are clustered at the year-2000 CZ level, and 95% confidence intervals are displayed. The sample includes individuals of the 1990-1994 birth cohorts observed in the 2005-2018 ACS and linked to themselves in the 2000 Long-Form Census; see text for details.